

# About us



Sean Stankowski

UKRI Future Leaders Fellow



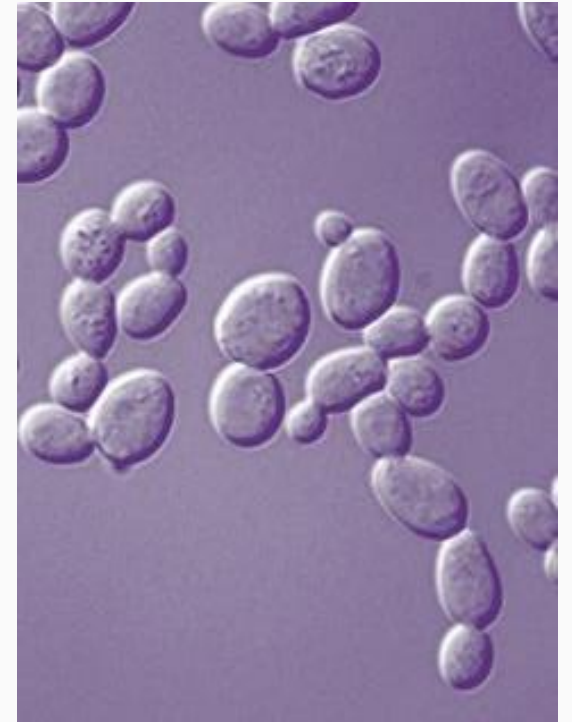
Tymoteusz Pieszko

Postdoctoral Fellow



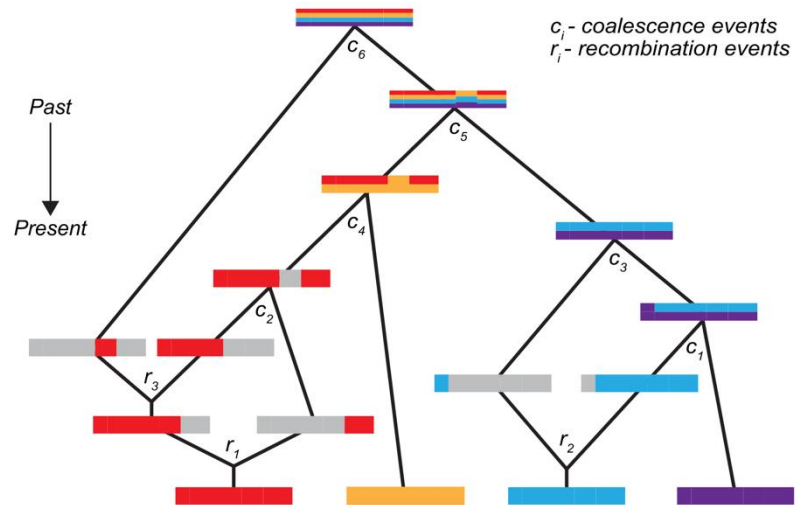
Department of Genetics, Evolution, and Environment

# We work with genomic data from diverse organisms



# Very interested in new methods

(b) Ancestral Recombination Graph

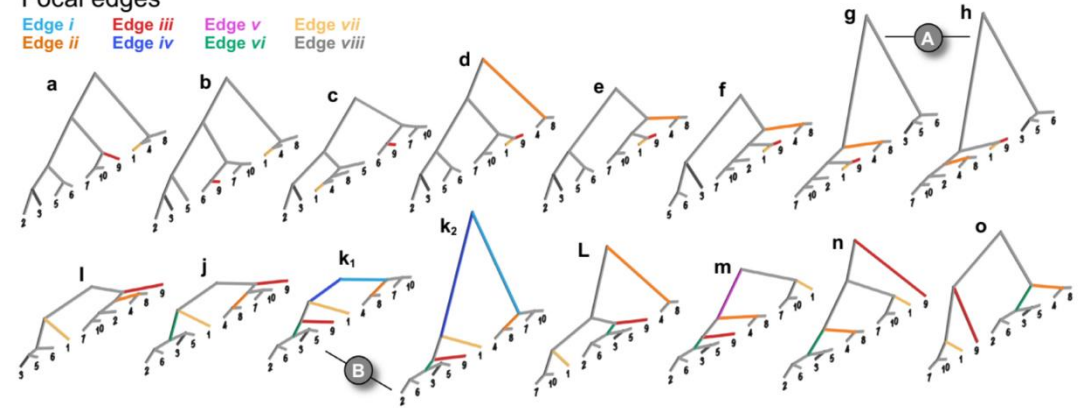


(a)

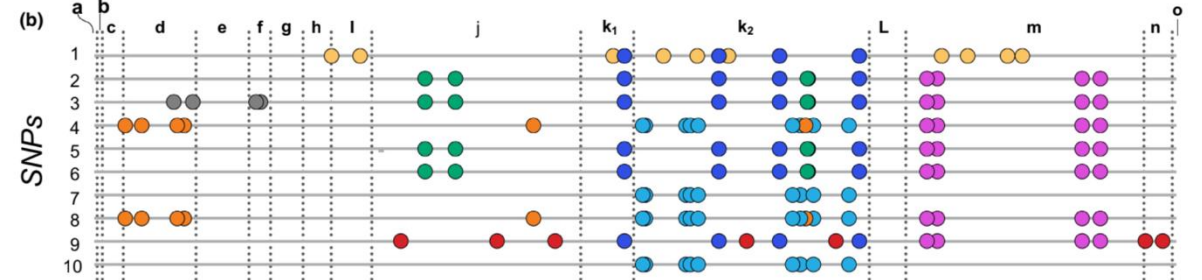
Focal edges

Edge i Edge iii Edge v Edge vii  
Edge ii Edge iv Edge vi Edge viii

Trees



(b)



## Motivated by questions about

- Evolutionary pathways to novelty
- Adaption and speciation
- Evolution of sex and asexuality
- Genome evolution

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- Develop confidence working in a high-performance computing environment.
- Provide practical experience with tools for read processing, mapping, variant calling, and filtering.
- Introduce analyses used to explore and assess genomic variation in a new dataset.

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- A comprehensive introduction to bioinformatics, Linux or programming.

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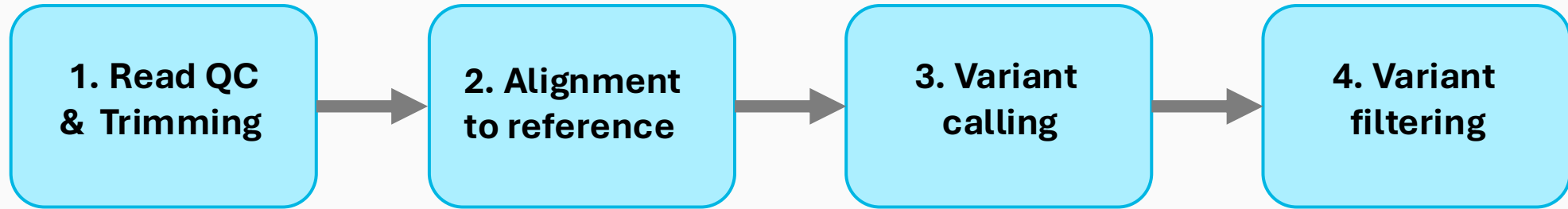
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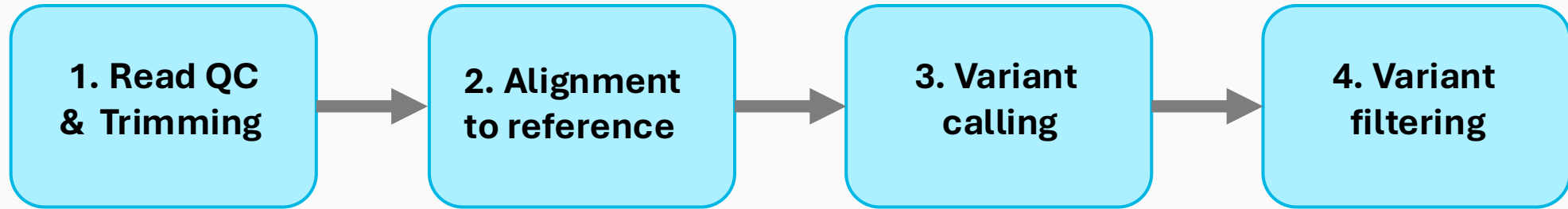
- A comprehensive introduction to bioinformatics, Linux or programming.
- A complete overview of all of methods and approaches that are available.
- A single “correct” or “best” pipeline that will be ideal for every dataset.
- A comprehensive guide to the best downstream genomic analysis for your research question.

# The plan for the next 2 days

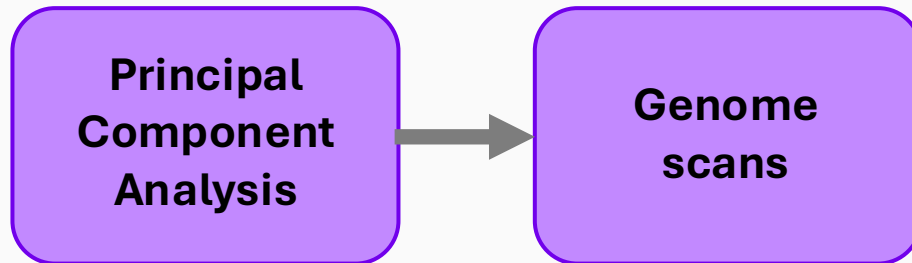


*Reads to SNPs*

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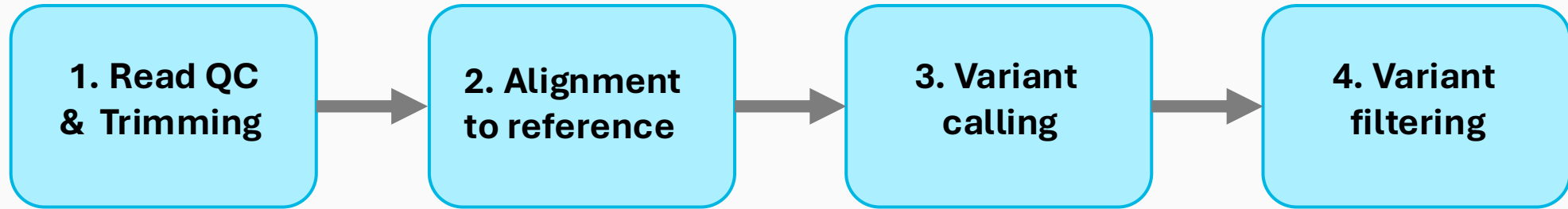


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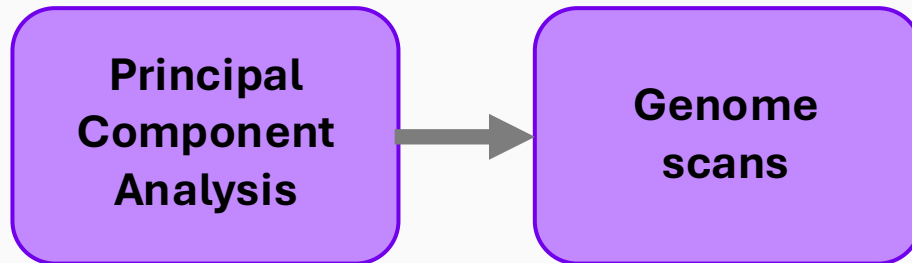


*Basic analysis*

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*Reads to SNPs*

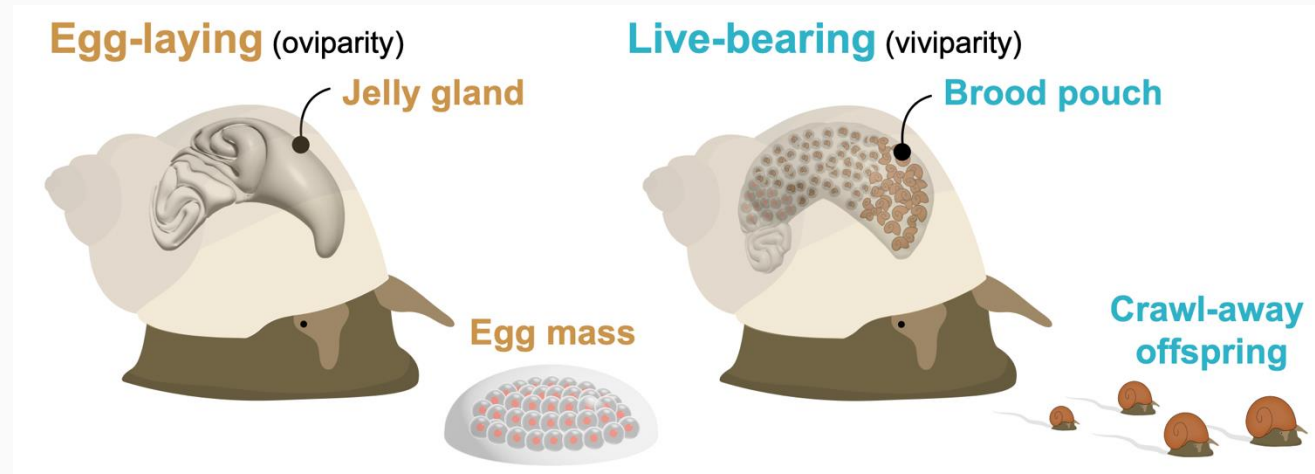


*Basic analysis*

**But we are at  
your service!**

# The dataset we will work with

- Intertidal snails from the genus *Littorina*
- Four taxa that differ in their reproductive mode
- 108 individuals Illumina sequenced to 15x coverage



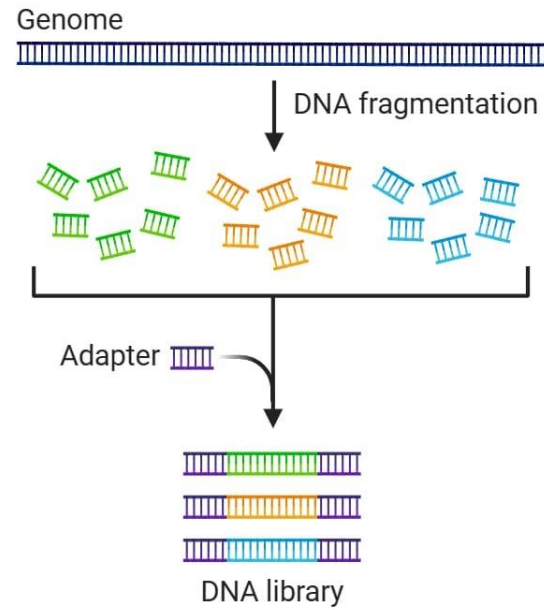
<https://www.science.org/doi/10.1126/science.adi2982>

**Background:**

Short-read sequencing and  
sequencing strategies

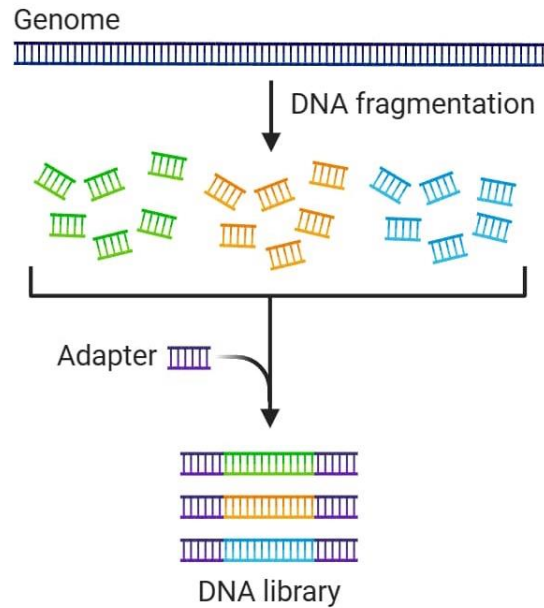
# Illumina sequencing

## ① Library preparation

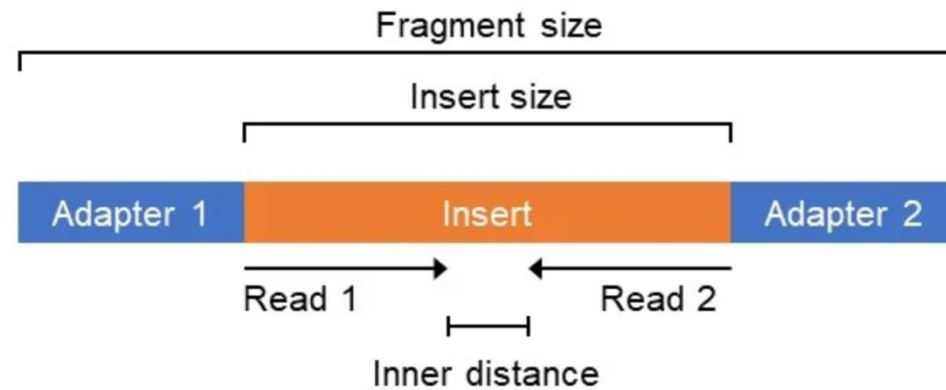


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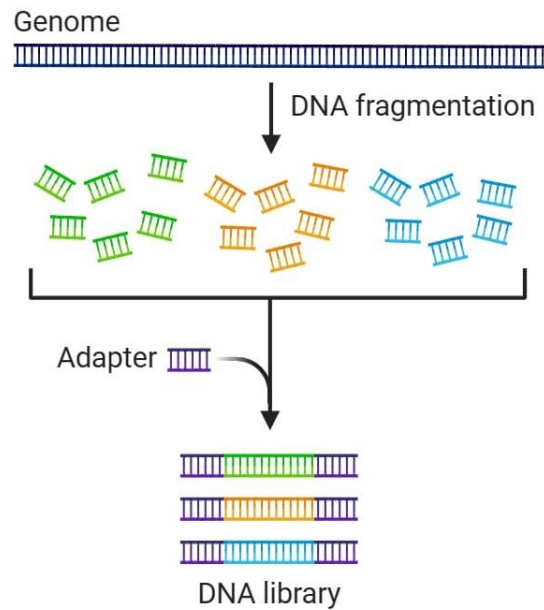


## Paired-end sequencing



# Illumina sequencing

## ① Library preparation



## NovaSeq X series specifications

|                      |                          |
|----------------------|--------------------------|
| Output Range         | ~165 Gb - 16 Tb          |
| Single reads per run | 1.6 billion - 52 billion |
| Read length          | 2 × 150 bp               |
| Run time             | ~13 hr - 48 hr           |

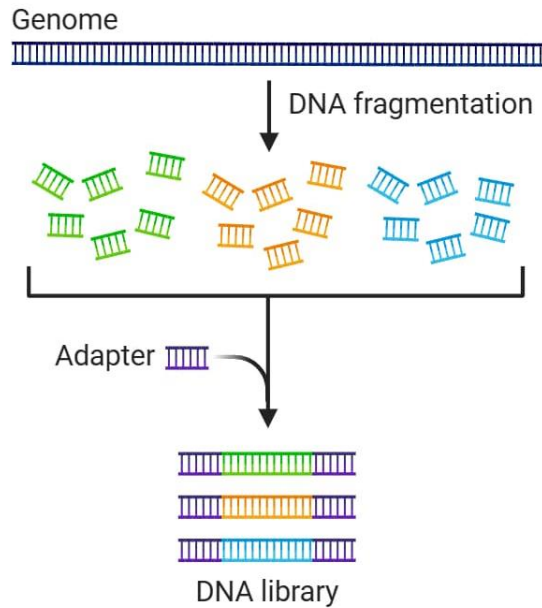
[View All NovaSeq X Specs](#)

 [View AR](#)

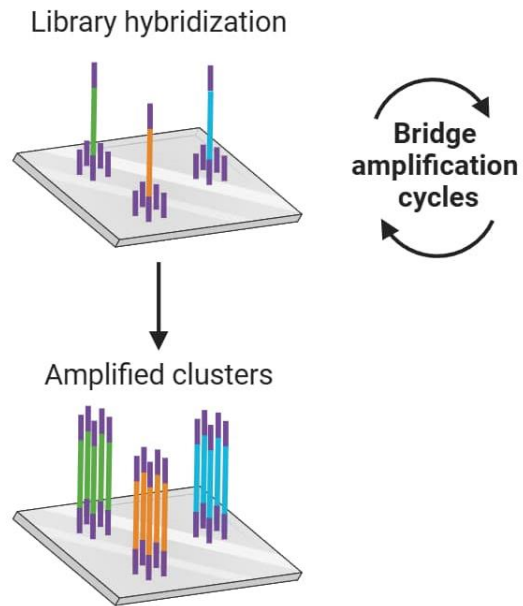
450 human genomes at 10x coverage per run

# Illumina sequencing

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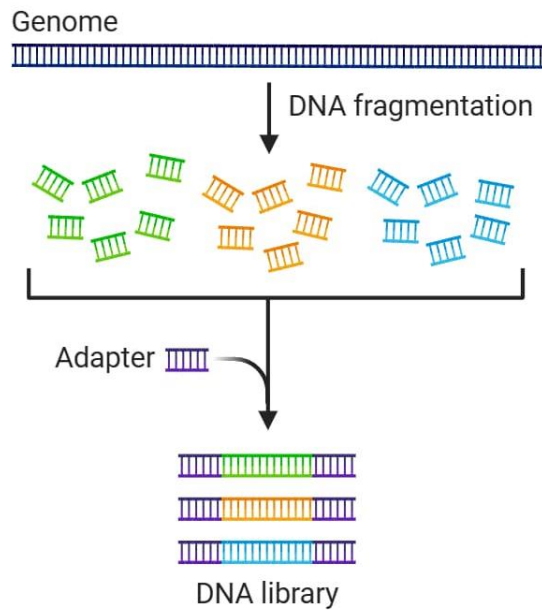


## ② DNA library bridge amplification

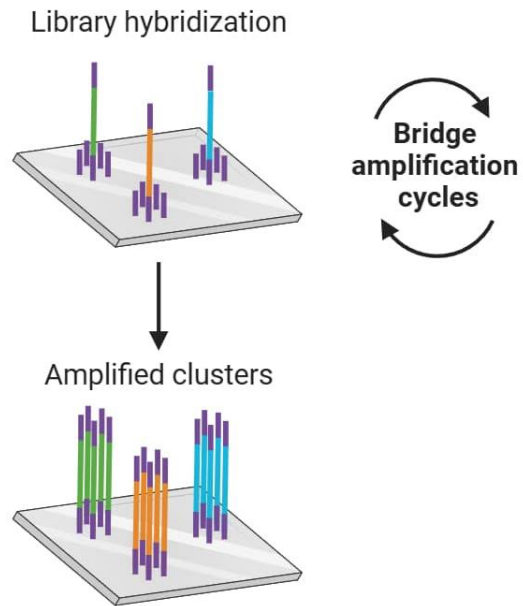


# Illumina sequencing

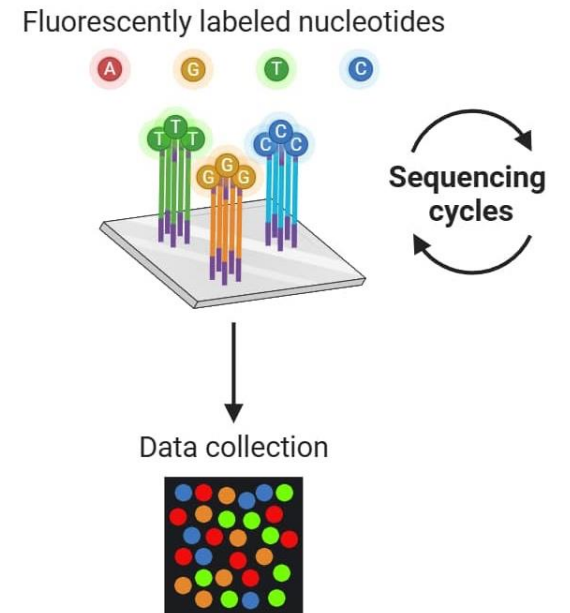
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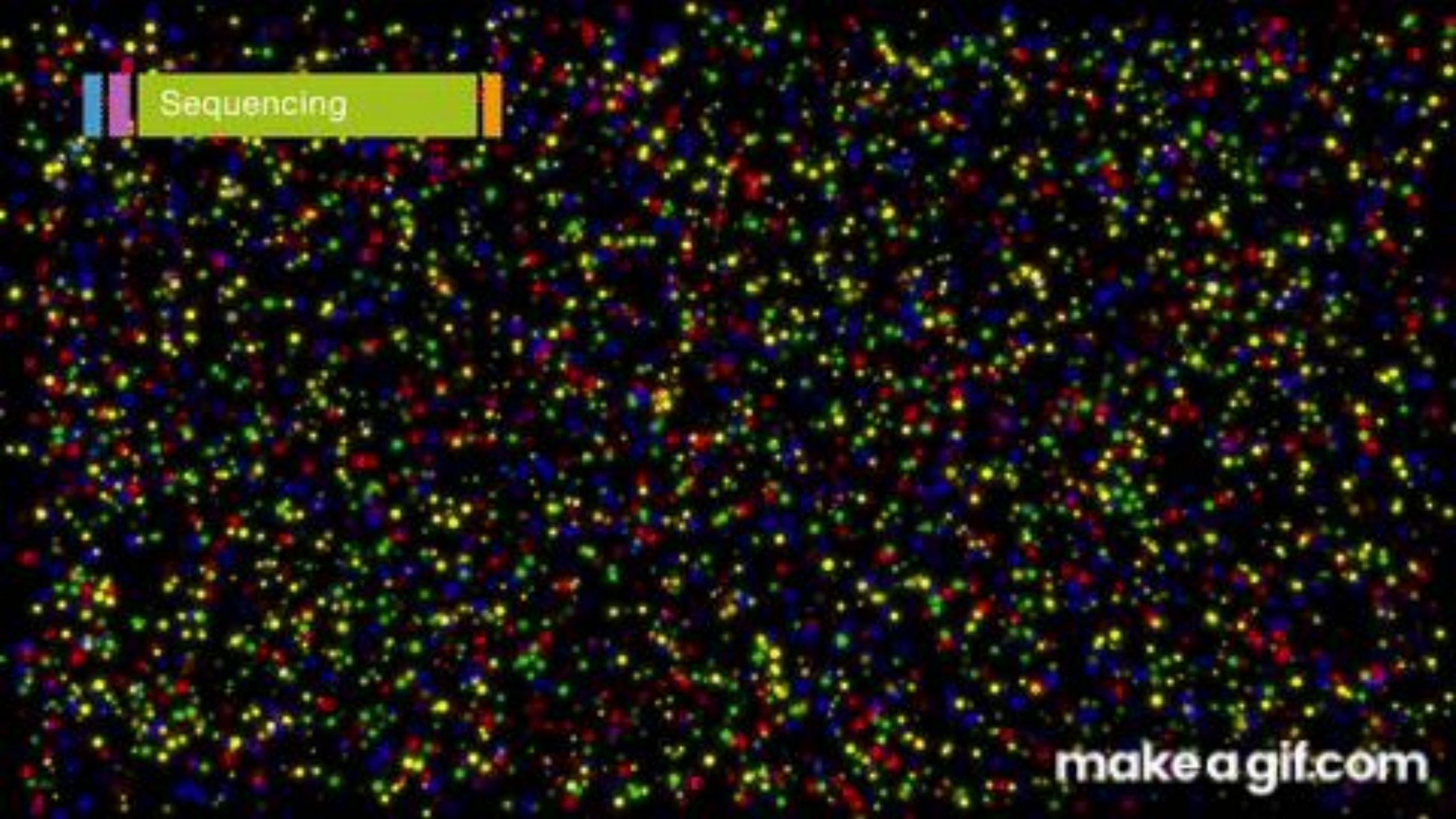


## ② DNA library bridge amplification



## ③ DNA library sequencing





Sequencing

makeagif.com

# Different sequencing designs

- **High-coverage WGS:** Sequences each individual's entire genome at sufficient depth to call genotypes accurately across most genomic regions.

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- **RADseq:** Sequences regions adjacent to restriction-enzyme cut sites, sampling a repeatable subset of the genome across many individuals.
- **Linked-read sequencing:** Uses barcoded short reads derived from the same long DNA molecules to recover long-range information, including haplotypes and structural variants.

# Different sequencing designs

| Approach          | Cost/sample | Sample size | Genome coverage       | Genotypes                 | Haplotypes                 | Main trade-off                           |
|-------------------|-------------|-------------|-----------------------|---------------------------|----------------------------|------------------------------------------|
| High-coverage WGS | High        | Low–medium  | Whole genome; deep    | High confidence           | Mostly statistical phasing | Very reliable calls, high cost           |
| Low-coverage WGS  | Low–medium  | High        | Whole genome; shallow | Likelihoods or imputation | Statistical phasing        | Broad coverage, less certainty           |
| Pool-seq          | Very low    | Very high   | Whole genome; pooled  | No individual genotypes   | None                       | Cheap allele frequencies, no individuals |
| RAD-seq           | Low         | High        | Small subset          | At sampled loci           | Local only                 | Cheap markers, most loci missed          |
| Linked reads      | Medium–high | Low–medium  | Whole genome          | High confidence           | Long-range phasing         | Strong phasing, specialised DNA/library  |

# Different sequencing designs

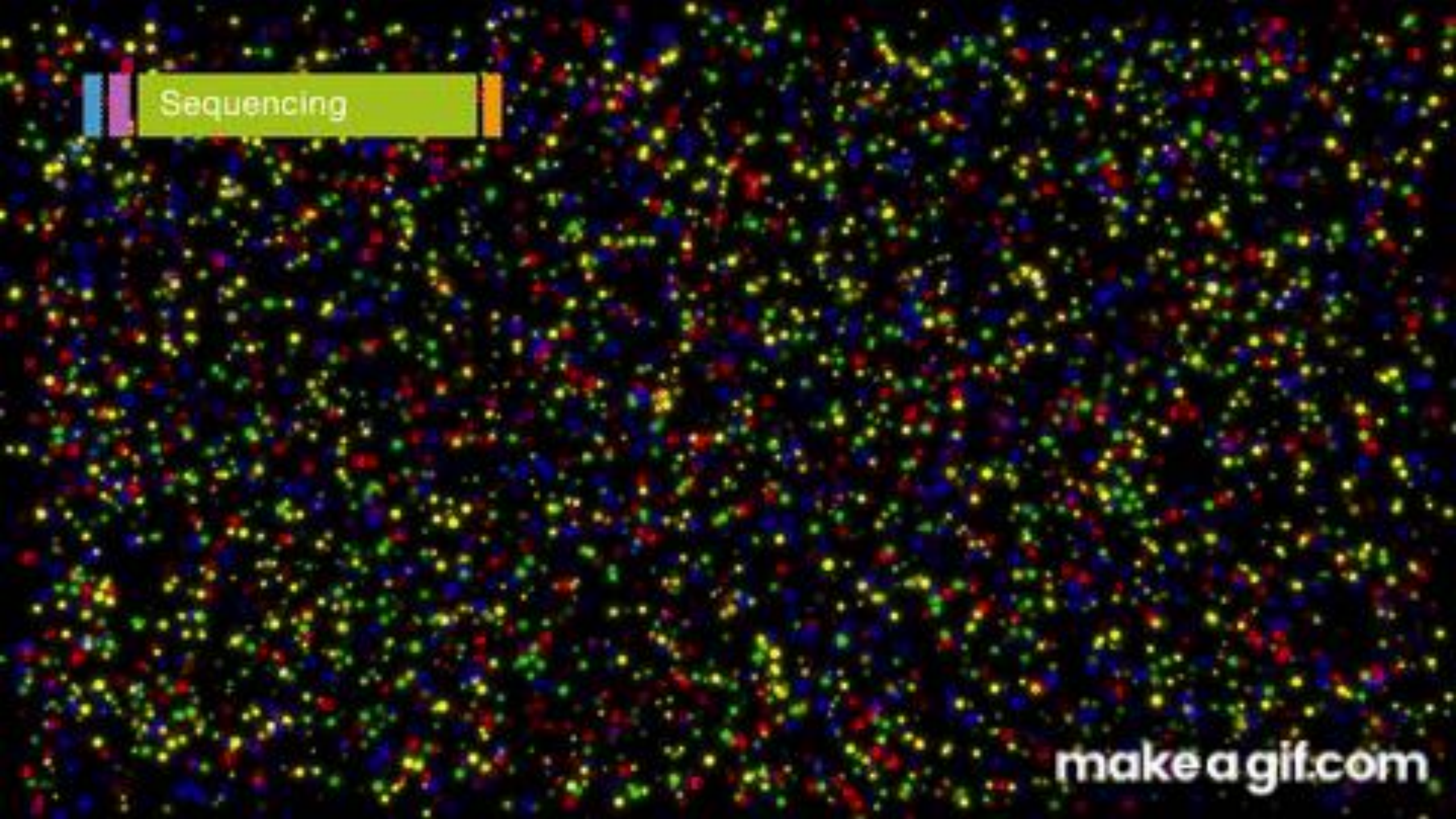
| Main objective                                                                    | Usually best suited approach |
|-----------------------------------------------------------------------------------|------------------------------|
| Most accurate genome-wide genotypes for each individual                           | High-coverage Illumina WGS   |
| Genome-wide population data from many individuals                                 | Low-coverage WGS             |
| Allele-frequency comparisons using the largest possible number of individuals     | Pool-seq                     |
| Population structure or phylogeography on a restricted budget                     | RAD-seq                      |
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*Questions?*

# **1.1 Quality control and trimming of raw short-read sequence data**



Sequencing

makeagif.com

# FASTQ Format

```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTCG
TGCAGCAGCTAATGAGGAACCACTTCCTCCCTCCAGCCGCTCTAAATACCTCAGACAATAGGATCATCATAATAATCCCCTAGTCTGAACTG
+
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAAGAAGAAAGCACAAAGTATTTACGCCTATCCTTCATATTTCCGCAAGGTAAGTATCTCGGTTTCATATCGAGATTTATATAGAATCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - FFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGGAAGTTATGCAGGGGCATCCTGTATTATTAAATAGAGCACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG
+
FF-88FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - F - FFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAGTACATCAACAAAAGAAACCTAACTGCCCTACCGGCAAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCCAGTCAATATAAGG
+
8F88FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - FFFFFFFFFF - FFFFFFFFFF - FFFFFFFFFF - FFFFFFFFFF
```

# FASTQ Format

[illegible]

@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979

# Header

GGAGGAAGGTCCTCGCTCCTCTTTCATATAAGGGAAATGGCTGAAT

## Sequence

 $+$ 

FFFFHHHHHJJJJJJJJJJJJGGIJJJJJJJJJJ

## Quality string

# FASTQ Header

## fastq header format (version > 1.8)

| Sequence Header |           |      |            |    |       |       | +Sequence ID |   |   |   |         |
|-----------------|-----------|------|------------|----|-------|-------|--------------|---|---|---|---------|
| a               | b         | c    | d          | e  | f     | g     | h            | i | j | k |         |
| @               | HWI-ST486 | :166 | :C06K9ACXX | :7 | :1101 | :1443 | :1995        | 1 | N | 0 | :ACAGTG |

**a. unique instrument name**

b. run id

c. flowcell id

d. flowcell lane

e. tile number within the flowcell lane

f. x-coordinate of the cluster within the tile

g. y-coordinate of the cluster within the tile

**h. the member of a pair, 1 or 2 (paired-end or mate-pair reads only)**

i. Y if the read fails filter (read is bad), N otherwise

j. 0 when no control bits are on

k. index sequence

# FASTQ Sequence

- Contains the nucleotide sequence generated by the sequencer.
- Bases are normally represented by A, C, G and T.
- N indicates that the sequencer could not confidently identify the base.
- The sequence is the read as produced by the instrument and **may still contain adapters or other technical sequence.**

Adapter

```
GTGCCAGCCGCCGCGGTAA|TACGAAGGGACCTAGCGTAGTTCGGAATTACTGGGCTTAAAGAGTTCGTAGGTGGTTAAAAAAGTTGGTGGTGAAATCCCAGAGCTTAACTCTGGAAGTGC  
CATCAAACTTTTTAGCTAGAGTATGATAGAGGAAAGCAGAATTTCTAGTGTAGAGGTGAAATTCGTAGATATTAGAAAGAATACCAATTGCGAAGGCAGCTTTCTGGATCATTACTGAC  
ACTGAGGAAC
```

# Phred+33 quality-scores encoding

| Phred score | Q0  | Q1  | Q2  | Q3  | Q4  | Q5  | Q6  | Q7  | Q8  | Q9  | Q10 | Q11 | Q12 | Q13 | Q14 | Q15 | Q16 | Q17 | Q18 | Q19 | Q20 |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Character   | !   | "   | #   | \$  | %   | &   | '   | (   | )   | *   | +   | ,   | -   | .   | /   | 0   | 1   | 2   | 3   | 4   | 5   |
| Phred score | Q21 | Q22 | Q23 | Q24 | Q25 | Q26 | Q27 | Q28 | Q29 | Q30 | Q31 | Q32 | Q33 | Q34 | Q35 | Q36 | Q37 | Q38 | Q39 | Q40 | Q41 |
| Character   | 6   | 7   | 8   | 9   | :   | ;   | <   | =   | >   | ?   | @   | A   | B   | C   | D   | E   | F   | G   | H   | I   | J   |

# Phred+33 quality-scores encoding

| Phred score | Q0  | Q1  | Q2  | Q3  | Q4  | Q5  | Q6  | Q7  | Q8  | Q9  | Q10 | Q11 | Q12 | Q13 | Q14 | Q15 | Q16 | Q17 | Q18 | Q19 | Q20 |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Character   | !   | "   | #   | \$  | %   | &   | '   | (   | )   | *   | +   | ,   | -   | .   | /   | 0   | 1   | 2   | 3   | 4   | 5   |
| Phred score | Q21 | Q22 | Q23 | Q24 | Q25 | Q26 | Q27 | Q28 | Q29 | Q30 | Q31 | Q32 | Q33 | Q34 | Q35 | Q36 | Q37 | Q38 | Q39 | Q40 | Q41 |
| Character   | 6   | 7   | 8   | 9   | :   | ;   | <   | =   | >   | ?   | @   | A   | B   | C   | D   | E   | F   | G   | H   | I   | J   |

| Phred score | Error probability |
|-------------|-------------------|
| Q10         | 1 in 10           |
| Q20         | 1 in 100          |
| Q30         | 1 in 1,000        |
| Q40         | 1 in 10,000       |

## Phred+33 quality-scores encoding

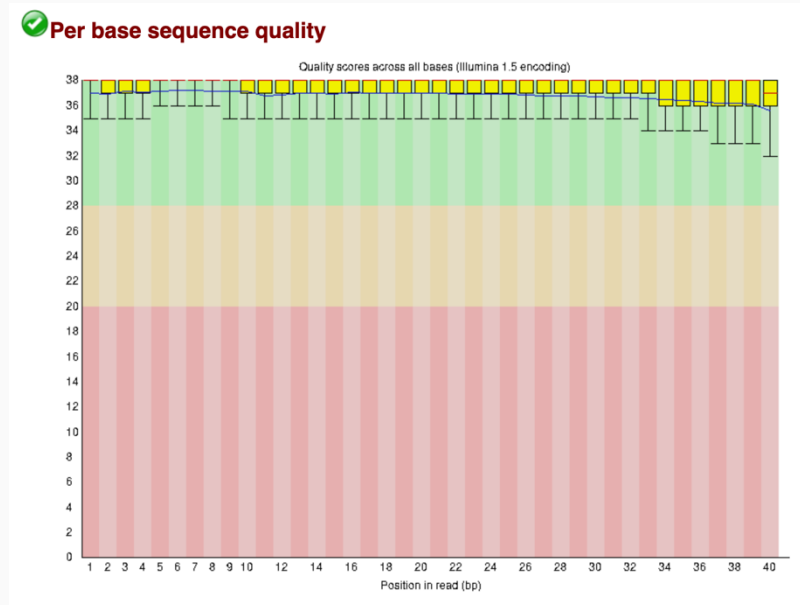
| Phred score | Q0  | Q1  | Q2  | Q3  | Q4  | Q5  | Q6  | Q7  | Q8  | Q9  | Q10 | Q11 | Q12 | Q13 | Q14 | Q15 | Q16 | Q17 | Q18 | Q19 | Q20 |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Character   | !   | "   | #   | \$  | %   | &   | '   | (   | )   | *   | +   | ,   | -   | .   | /   | 0   | 1   | 2   | 3   | 4   | 5   |
| Phred score | Q21 | Q22 | Q23 | Q24 | Q25 | Q26 | Q27 | Q28 | Q29 | Q30 | Q31 | Q32 | Q33 | Q34 | Q35 | Q36 | Q37 | Q38 | Q39 | Q40 | Q41 |
| Character   | 6   | 7   | 8   | 9   | :   | ;   | <   | =   | >   | ?   | @   | A   | B   | C   | D   | E   | F   | G   | H   | I   | J   |

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```
@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979
GGAGGAAGGTCCTCGCTCCTCTTTCATATAAGGGAAATGGCTGAAT
+
FFFFHHHHHHHJIIJJJJJJJJIIJJJIGIGIGGIJJIIJIIJJJJJJIIII
```

# What do we need to do with these raw data?

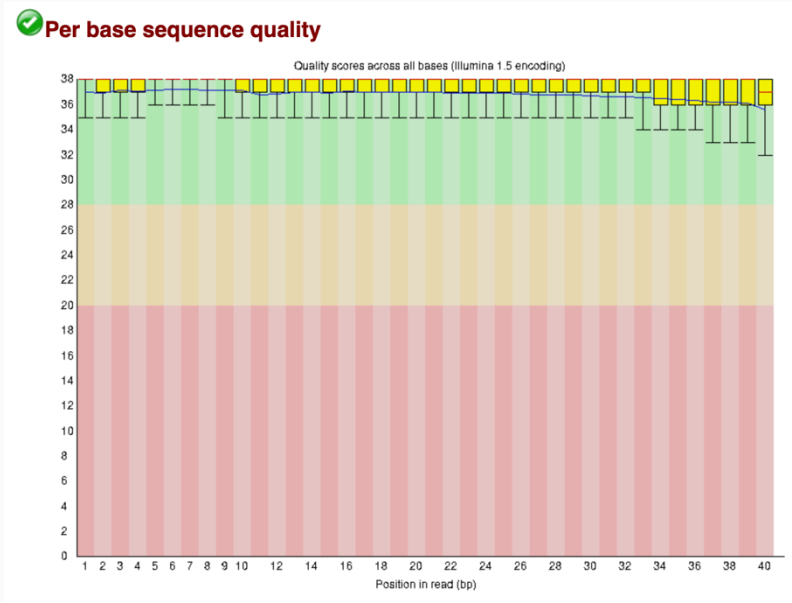
- Preform initial QC
  - Check the length and quality of reads etc...



Nice per base quality!

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Nice per base quality!



Not so nice...

# What do we need to do with these raw data?

- Preform initial QC
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- Trim non-biological sequences and discard low-quality reads



```
NNNNNTTTTTTTTTTNGATCNNNNNAAAAAAAAANNNNNN
#####&&&&&&&&+++++00000+++++#####&&&&&
```

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- Preform initial QC
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- Re-run QC to evaluate outcome

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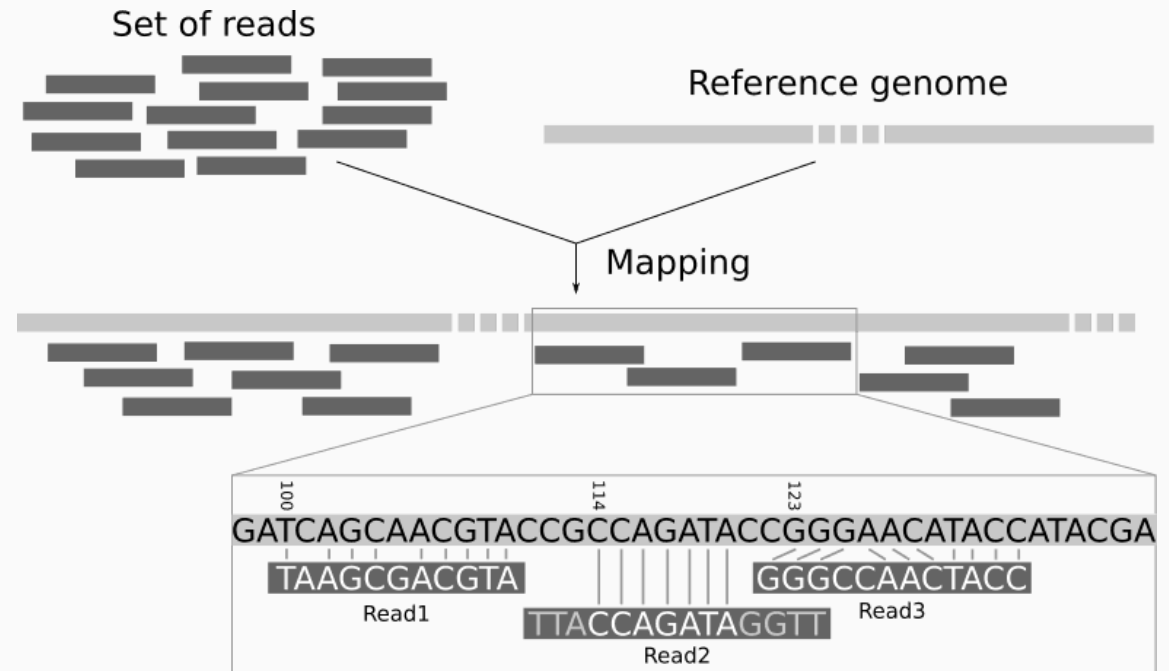
**Let's practice**

## **1.2 Mapping short-read data to a reference genome**

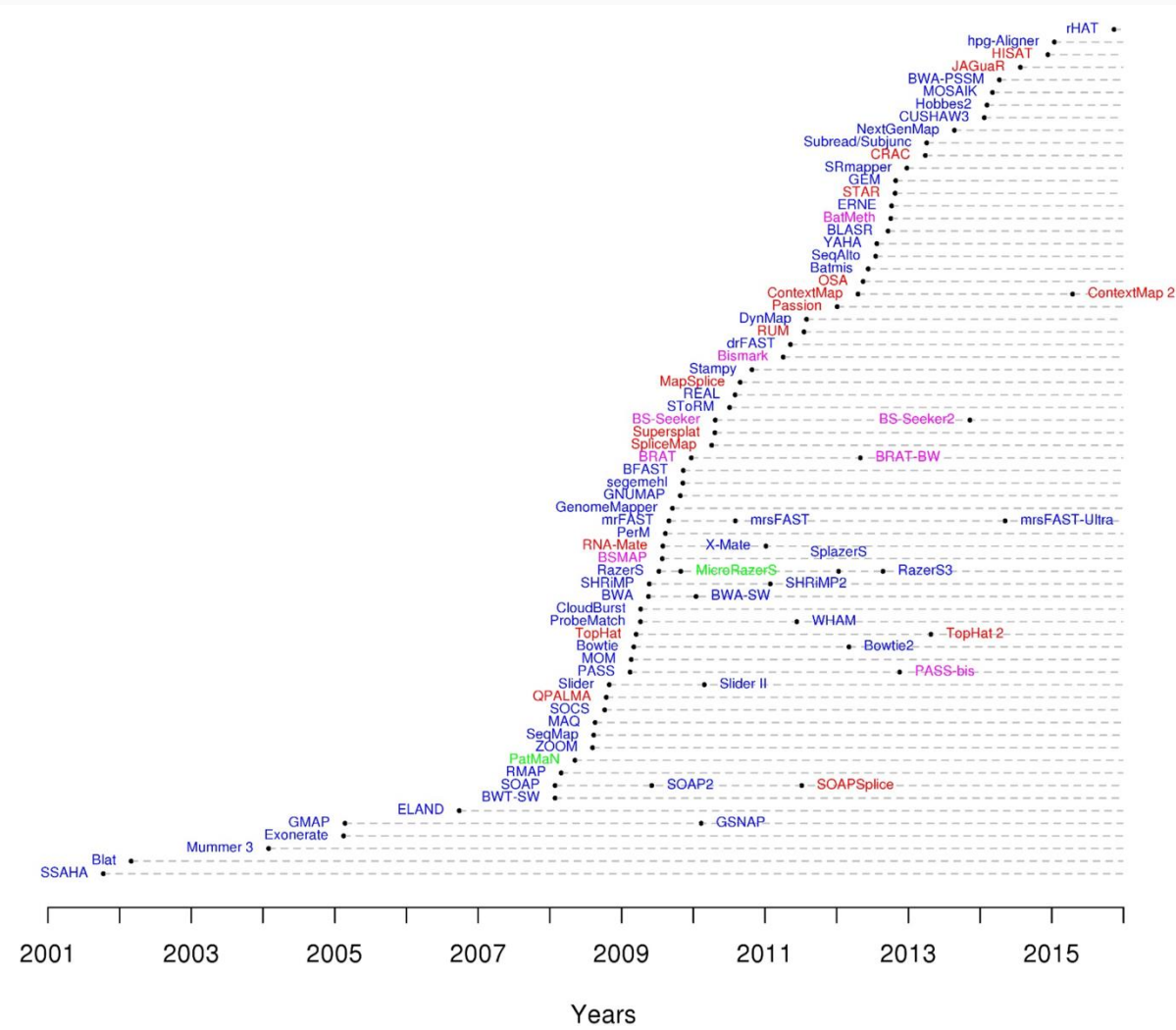
# What is read mapping?

The process of finding where a sequencing read most likely originated within a reference genome

<https://training.galaxyproject.org/training-material/topics/sequence-analysis/tutorials/mapping/tutorial.html>



# More than 90 mappers to choose from!



<https://www.ecseq.com/support/ngs/what-is-the-best-ngs-alignment-software>

# BWA-MEM2

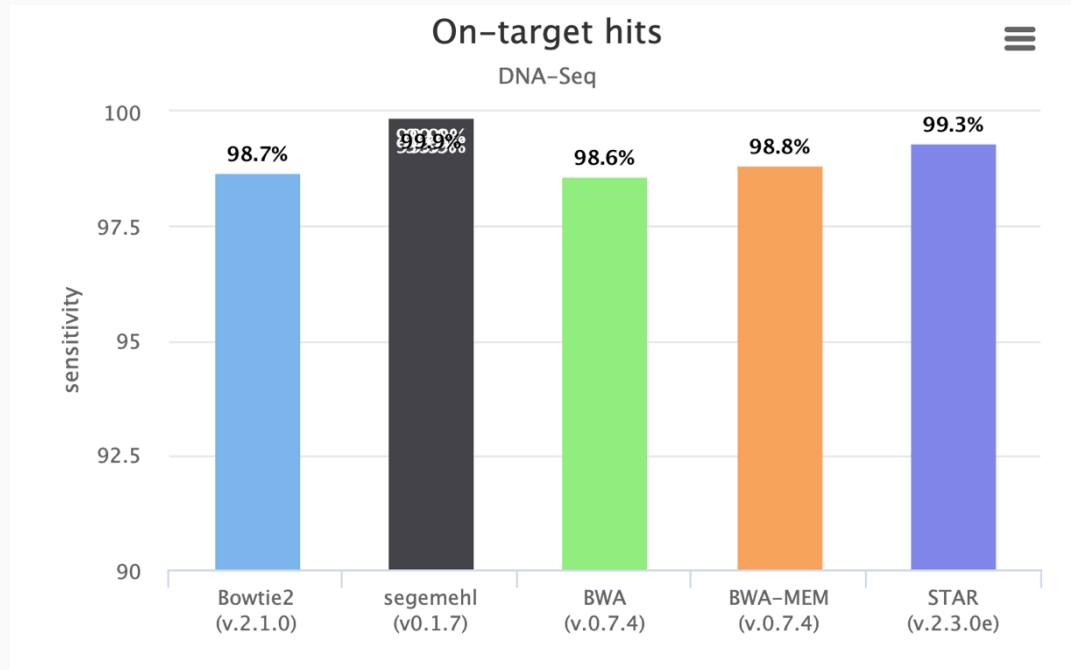
<https://github.com/bwa-mem2/bwa-mem2>

- Very widely used, efficient, and open source
- Leverages the Burrows-Wheeler transform
- The **Burrows–Wheeler transform (BWT)** makes short-read mapping efficient by reorganizing the reference genome into an index that can be **searched rapidly**.

[https://en.wikipedia.org/wiki/Burrows%E2%80%93Wheeler\\_transform](https://en.wikipedia.org/wiki/Burrows%E2%80%93Wheeler_transform)

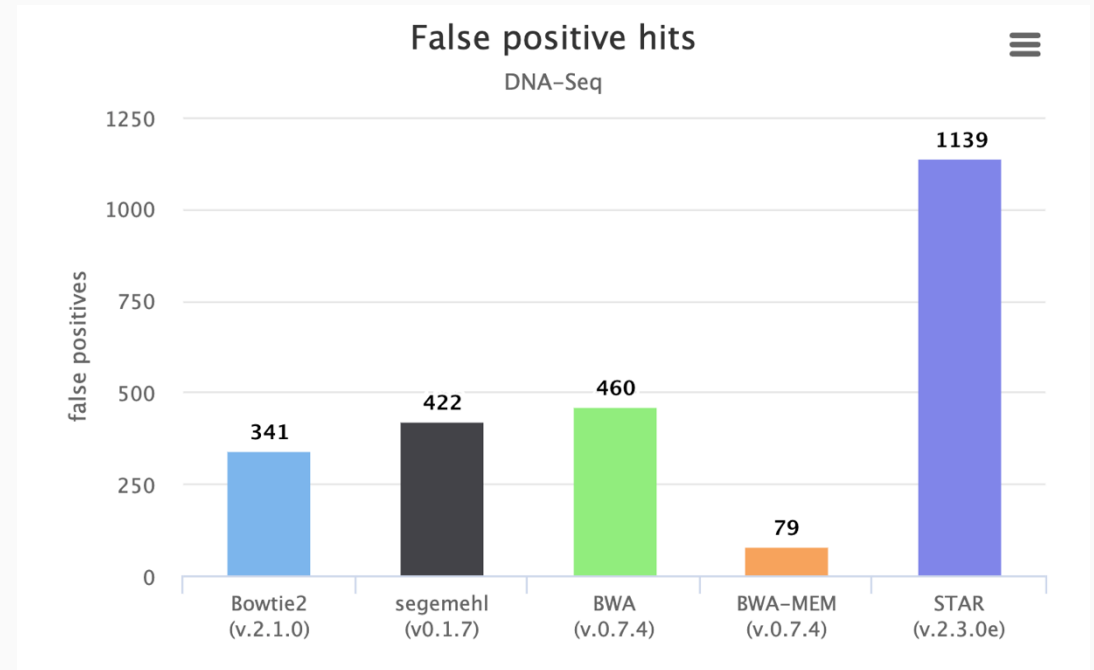
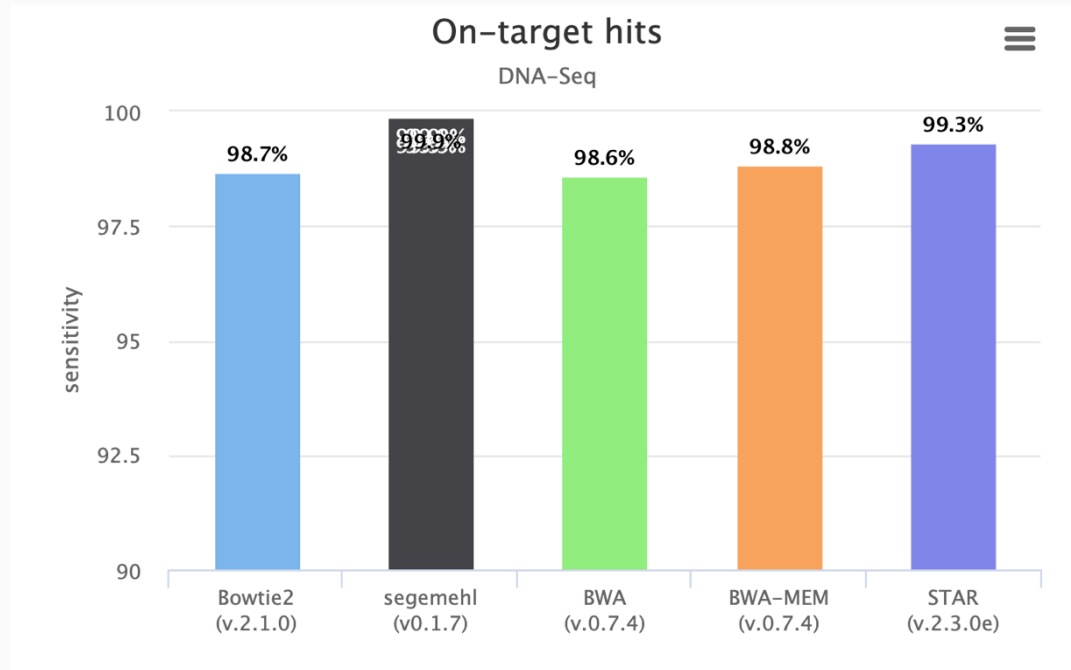
<https://www.youtube.com/watch?v=4n7NPk5lwbl>

# BWA-MEM2: performance



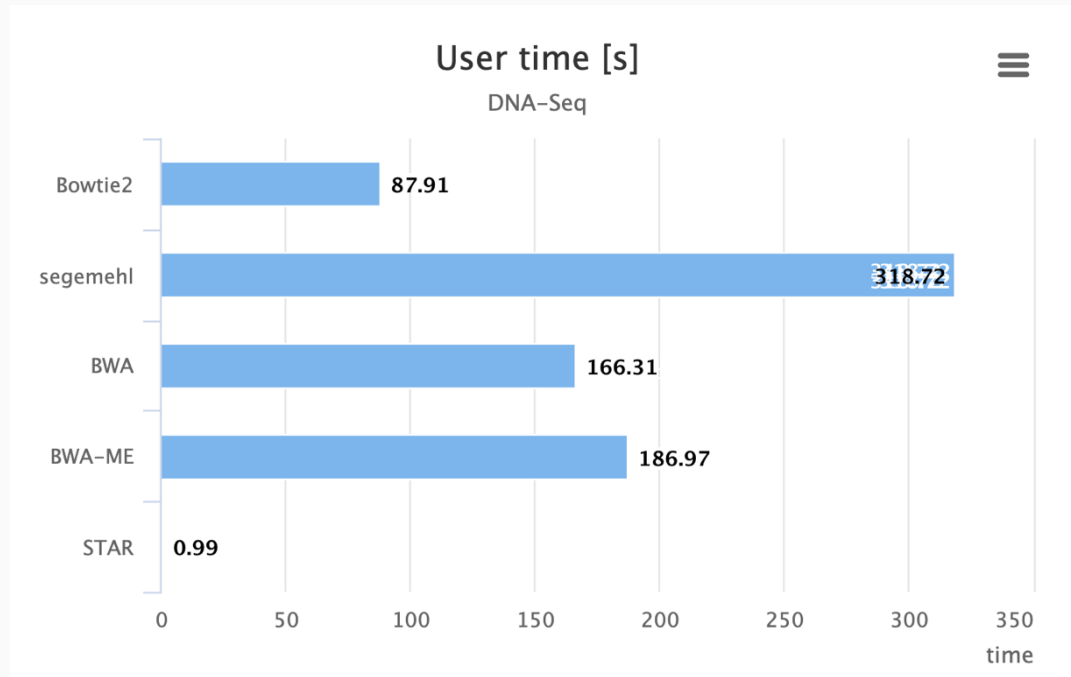
<https://www.ecseq.com/support/benchmark>

# BWA-MEM2: performance



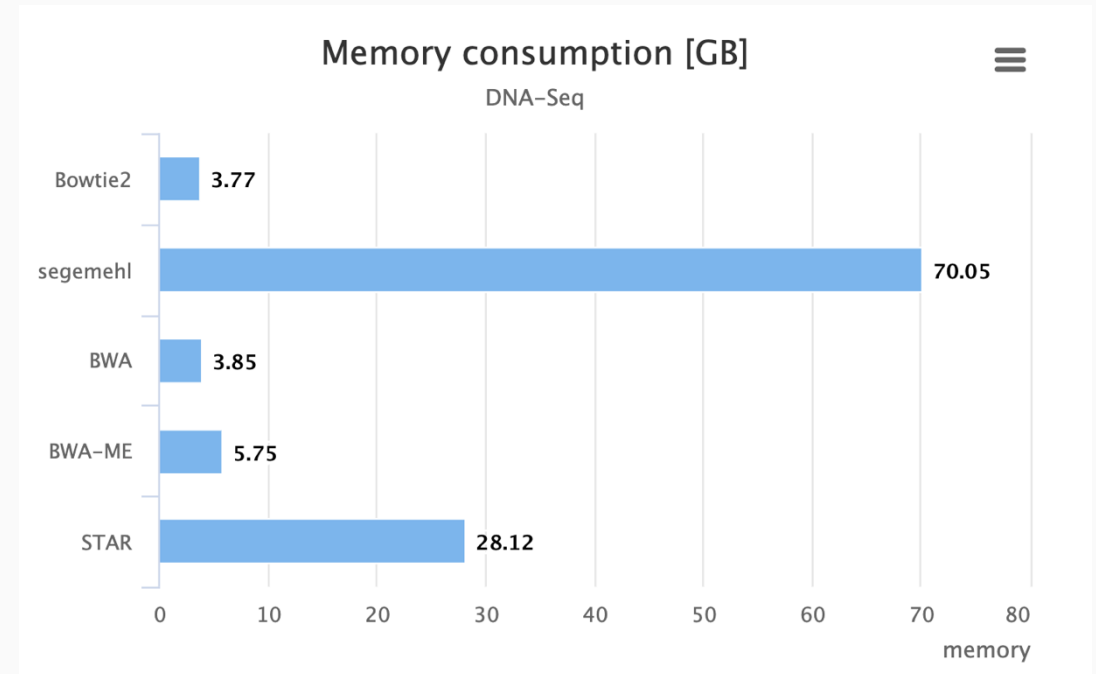
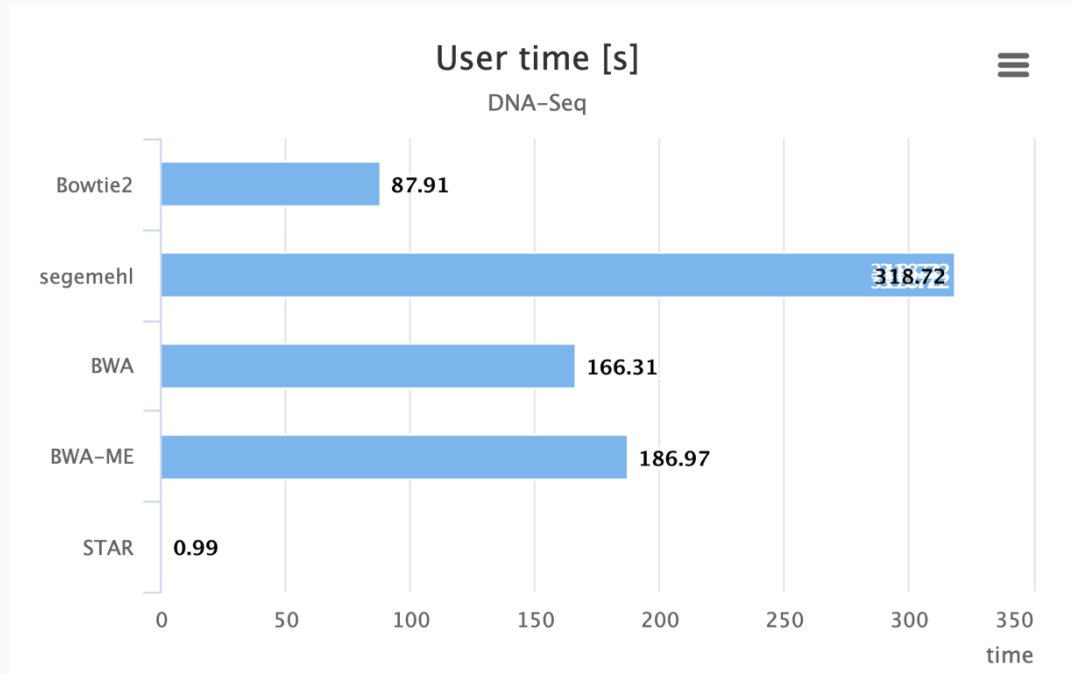
<https://www.ecseq.com/support/benchmark>

# BWA-MEM2: performance



<https://www.ecseq.com/support/benchmark>

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# SAM and BAM files

- **A SAM file (Sequence Alignment/Map)** is a human-readable, tab-delimited text file that records how sequencing reads align to a reference genome.
- Contains an optional header followed by one line for each read alignment, including its genomic position, mapping quality, CIGAR string, sequence and quality scores.

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|                          |      |     |    |    |            |   |    |     |                   |                            |                   |
|--------------------------|------|-----|----|----|------------|---|----|-----|-------------------|----------------------------|-------------------|
| @HD VN:1.5 SO:coordinate |      |     |    |    |            |   |    |     |                   |                            | Header section    |
| @SQ SN:ref LN:45         |      |     |    |    |            |   |    |     |                   |                            |                   |
| r001                     | 99   | ref | 7  | 30 | 8M2I4M1D3M | = | 37 | 39  | TTAGATAAAGGATACTG | *                          | Alignment section |
| r002                     | 0    | ref | 9  | 30 | 3S6M1P1I4M | * | 0  | 0   | AAAAGATAAGGATA    | *                          |                   |
| r003                     | 0    | ref | 9  | 30 | 5S6M       | * | 0  | 0   | GCCTAAGCTAA       | * SA:Z:ref,29,-,6H5M,17,0; |                   |
| r004                     | 0    | ref | 16 | 30 | 6M14N5M    | * | 0  | 0   | ATAGCTTCAGC       | *                          |                   |
| r003                     | 2064 | ref | 29 | 17 | 6H5M       | * | 0  | 0   | TAGGC             | * SA:Z:ref,9,+,5S6M,30,1;  |                   |
| r001                     | 147  | ref | 37 | 30 | 9M         | = | 7  | -39 | CAGCGGCAT         | * NM:i:1                   |                   |

Optional fields in the format of TAG:TYPE:VALUE

QUAL: read quality; \* meaning such information is not available

SEQ: read sequence

TLEN: the number of bases covered by the reads from the same fragment. Plus/minus means the current read is the leftmost/rightmost read. E.g. compare first and last lines.

PNEXT: Position of the primary alignment of the NEXT read in the template. Set as 0 when the information is unavailable. It corresponds to POS column.

RNEXT: reference sequence name of the primary alignment of the NEXT read. For paired-end sequencing, NEXT read is the paired read, corresponding to the RNAME column.

CIGAR: summary of alignment, e.g. insertion, deletion

MAPQ: mapping quality

POS: 1-based position

RNAME: reference sequence name, e.g. chromosome/transcript id

FLAG: indicates alignment information about the read, e.g. paired, aligned, etc.

QNAME: query template name, aka. read ID

# SAM and BAM files

- A **SAM file (Sequence Alignment/Map)** is a human-readable, tab-delimited text file that records how sequencing reads align to a reference genome.
- Contains an optional header followed by one line for each read alignment, including its genomic position, mapping quality, CIGAR string, sequence and quality scores.

|                          |      |     |    |    |            |   |    |     |                   |                            |                   |
|--------------------------|------|-----|----|----|------------|---|----|-----|-------------------|----------------------------|-------------------|
| @HD VN:1.5 SO:coordinate |      |     |    |    |            |   |    |     |                   |                            | Header section    |
| @SQ SN:ref LN:45         |      |     |    |    |            |   |    |     |                   |                            |                   |
| r001                     | 99   | ref | 7  | 30 | 8M2I4M1D3M | = | 37 | 39  | TTAGATAAAGGATACTG | *                          | Alignment section |
| r002                     | 0    | ref | 9  | 30 | 3S6M1P1I4M | * | 0  | 0   | AAAAGATAAGGATA    | *                          |                   |
| r003                     | 0    | ref | 9  | 30 | 5S6M       | * | 0  | 0   | GCCTAAGCTAA       | * SA:Z:ref,29,-,6H5M,17,0; |                   |
| r004                     | 0    | ref | 16 | 30 | 6M14N5M    | * | 0  | 0   | ATAGCTTCAGC       | *                          |                   |
| r003                     | 2064 | ref | 29 | 17 | 6H5M       | * | 0  | 0   | TAGGC             | * SA:Z:ref,9,+,5S6M,30,1;  |                   |
| r001                     | 147  | ref | 37 | 30 | 9M         | = | 7  | -39 | CAGCGGCAT         | * NM:i:1                   |                   |

<

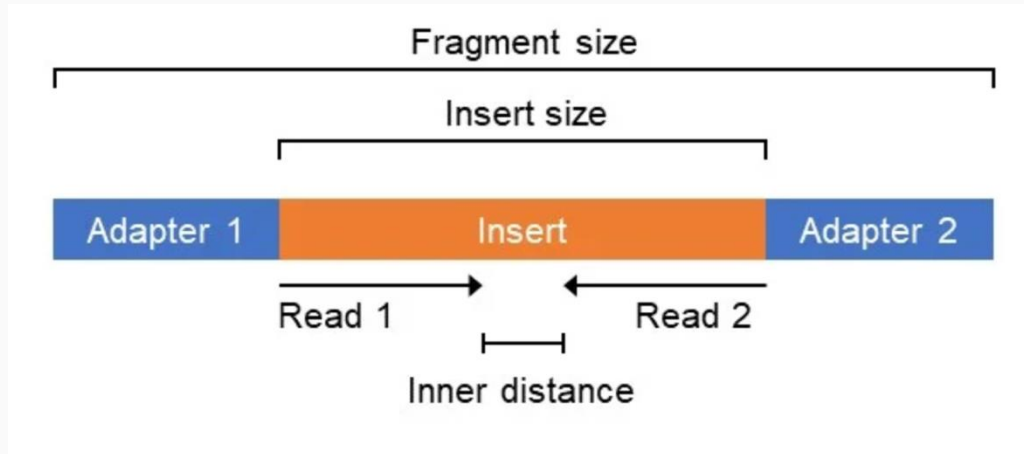
## BAM (Binary Alignment/Map)

contains the same information but stored in a compressed binary format.

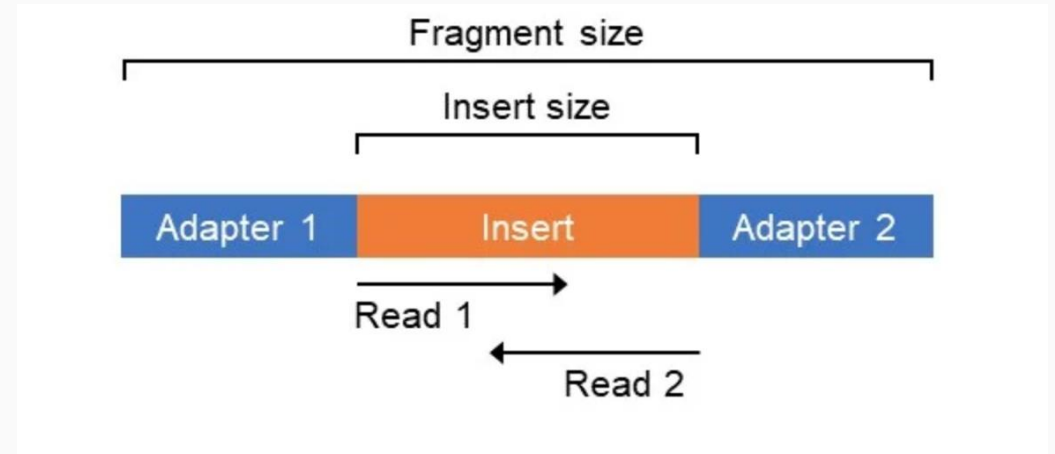
BAM files are smaller, faster for software to process and can be indexed, allowing programs to retrieve reads from a particular genomic region without reading the entire file.

# Most Illumina sequences are now paired-end

Non-overlapping PE



Overlapping PE

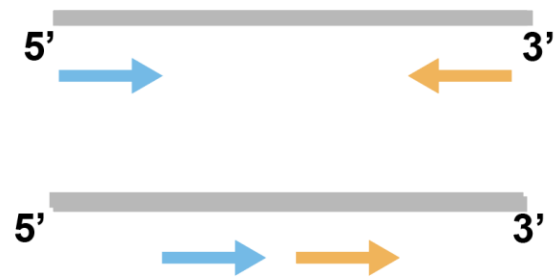


# Paired reads allow us to assess mapping

Properly Paired Reads (PPR)



Improperly Paired Reads (IPR)



Mates map on different scaffolds



One of the mates does not map

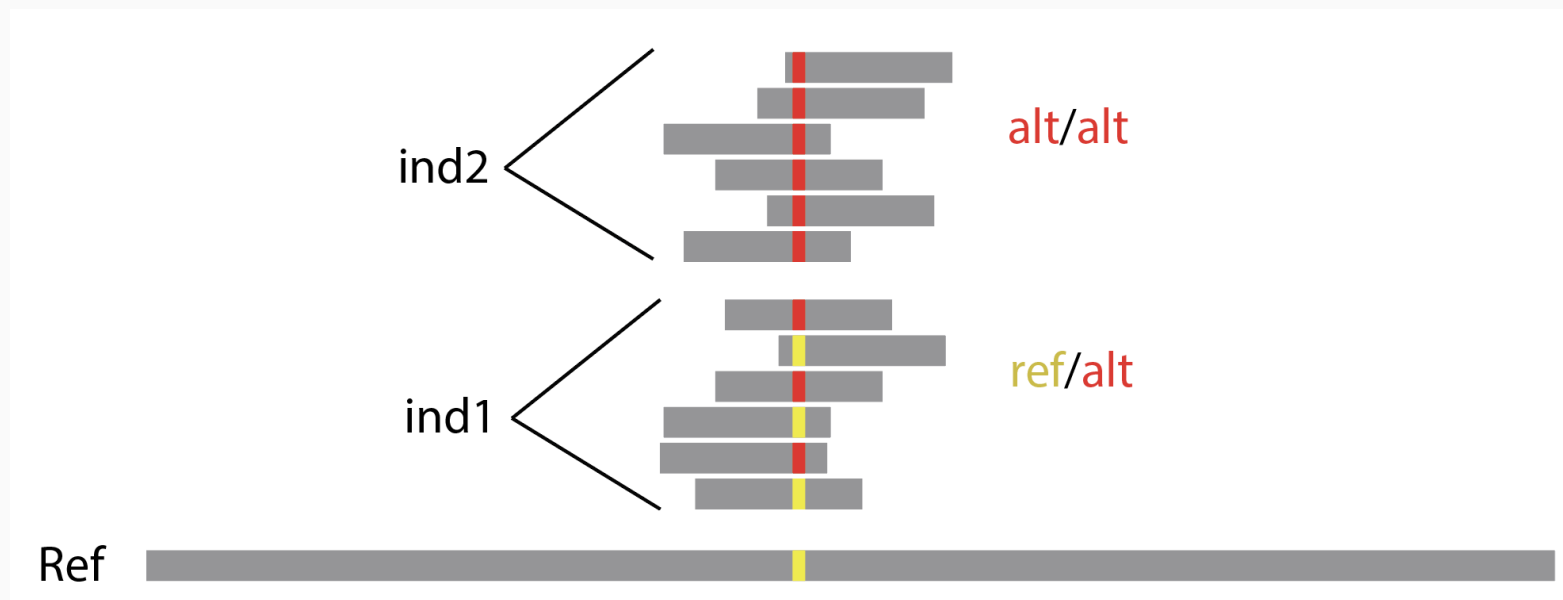


## **1.3 Multi-sample variant calling with BCFtools**

# Variant calling

The process of inferring genetic differences from sequencing reads relative to a reference genome:

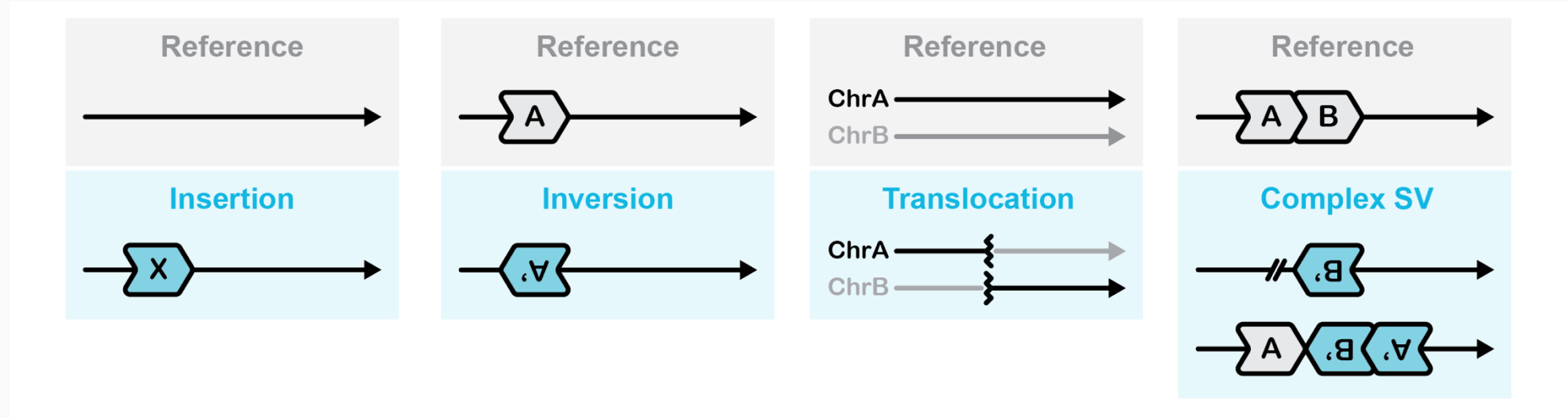
A single-nucleotide polymorphism (SNP)



# Variant calling

The process of inferring genetic differences from sequencing reads relative to a reference genome:

Structural variants also!



# Variant calling

Each individual carries 2 alleles, and we want to infer the state of both from  $n$  reads.

# Variant calling

Each individual carries 2 alleles, and we want to infer the state of both from  $n$  reads.



How random flips would you want to confidently conclude that this is not a 2-headed coin?

*The reality is that you can never be 100% certain*

# Variant calling

Each individual carries 2 alleles, and we want to infer the state of both from  $n$  reads.

$$P_{\text{both alleles}} = 1 - 2^{1-n}$$

where  $n$  is the number of reads

```
reads <- 1:20
prob_both <- 1 - 2^(1 - reads)
data.frame(
  reads = reads,
  probability = prob_both,
  percentage = 100 * prob_both
)
```

| Reads | Probability of observing both alleles |
|-------|---------------------------------------|
| 1     | 0.00%                                 |
| 2     | 50.00%                                |
| 3     | 75.00%                                |
| 4     | 87.50%                                |
| 5     | 93.75%                                |
| 6     | 96.88%                                |
| 7     | 98.44%                                |
| 8     | 99.22%                                |
| 9     | 99.61%                                |
| 10    | 99.80%                                |
| 11    | 99.90%                                |
| 12    | 99.95%                                |
| 13    | 99.98%                                |
| 14    | 99.99%                                |
| 15    | 99.994%                               |

# Variant calling

**How many genotypes would we would expect to miss-call with 5x coverage?**

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@ 5x coverage, we have a 93.75% chance of observing both alleles

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**How many genotypes would we would expect to miss-call with 5x coverage?**

@ 5x coverage, we have a 93.75% chance of observing both alleles

Genome-wide heterozygosity in *Littorina* is ~0.5% and genome size is 1.35 gbp, so we expect 6.5 million heterozygous calls per individual

$$(1-93.75) * 6,500,000 = 406,250...$$

# Variant calling can be informed by the ‘population’

Hardy–Weinberg equilibrium can give a prior expectation for genotype frequencies when variants are called jointly across individuals.

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$$P(AA) = (1 - p)^2, \quad P(Aa) = 2p(1 - p), \quad P(aa) = p^2$$

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- If the estimated alternative allele frequency is  $p$ , then

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- The caller combines the evidence from the sequencing reads with the expected frequency of each genotype to estimate which genotype is most likely.
- The effect of the prior is strongly dependent on the number of reads that support a genotype.

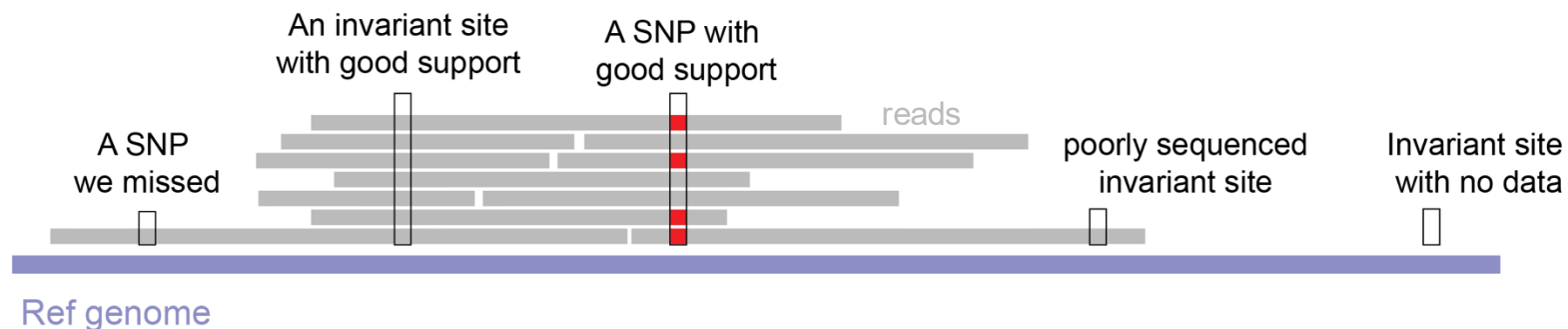
# Its not all about variants!

- The default for most callers is to only write variant sites to the output
- Many analyses require information from both variant and invariant sites

| #CHROM  | POS | Ind1 | Ind2 | Ind3 | Ind4 | Ind5 | Ind6 |
|---------|-----|------|------|------|------|------|------|
| Contig1 | 100 | 0/0  | 0/0  | 0/1  | 0/1  | 1/1  | 1/1  |
| Contig1 | 110 | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  | 1/1  |

# Its not all about variants!

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- Many analyses require information from both variant and invariant sites
- We cannot assume that sites not called as variants are invariant
- e.g., a site not in our variant-only VCF may have been poorly sequenced, rather than showing strong evidence for being monomorphic



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- The default for most callers is to only write variant sites to the output
- Many analyses require information from both variant and invariant sites
- We cannot assume that sites not called as variants are invariant
- e.g., a site not in our variant-only VCF may have been poorly sequenced, rather than showing strong evidence for being monomorphic
- The same level of evidence must be used for identifying variant or invariant sites
- For many caller there is an option to emit all sites (i.e., not just variants)—use this or live to regret it!

# Many callers are available!

They have very different philosophies!

- **Pileup callers:** Evaluate each site independently using the reads aligned to it; fast but less effective for complex variation.
- **Haplotype callers:** Reconstruct local sequences to identify SNPs, indels and complex variants.
- **Machine-learning callers:** Learn to distinguish variants from errors using validated training data.
- **Genotype-likelihood approaches:** Retain uncertainty in each genotype rather than making hard calls.

# Many callers are available!

They have very different philosophies!

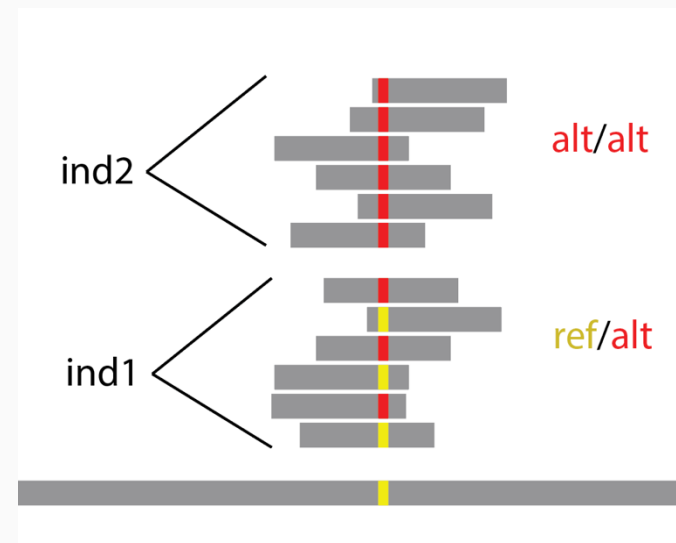
- **Pileup callers:** Evaluate each site independently using the reads aligned to it; fast but less effective for complex variation.
- **Haplotype callers:** Reconstruct local sequences to identify SNPs, indels and complex variants.

**BCFtools**  
mpileup | call



# BCFtools mpileup | call

- **Each base called independently of all others**
- **Builds a pileup:** Collects reads aligned to each position in the reference genome.
- **Weights read evidence:** Uses base and mapping qualities rather than simply counting alleles.
- **Calculates genotype likelihoods:** Estimates how well the reads support each possible genotype.
- **Calls variants across samples:** By default, treats samples as one population and applies an HWE assumption (can be deactivated using `-G` option)





- **Reconstructs haplotypes:** Locally reassembles reads into candidate sequences rather than examining sites independently.
- **Calculates genotype likelihoods:** Uses a Pair-HMM to determine how well each haplotype explains the reads.
- **Supports joint genotyping:** Calls samples individually as gVCFs before combining evidence across samples.
- **Much easier to combined new samples with old ones:** with others all individuals need to be recalled

```
A T G T T G G T A G C T A G T T T A G C
A A G C T A G T A G C T A A T T T T G C
A A G C T A G T A G C T A A T T T T G C
A T G T T G G T A G C T A G T T T A G C
A A G C T N G T A G C T A A T T T T G C
A A G C T A G T A G C T A A T T T T G C
A T G T T G G T A G C T A G T T T A G C
A T G T T G G T A G C T A G T T T A G C
A T G T T G G T A G C T A G T T T A G C
A A G C T A G T A G C T A A T T T N G C
A A G C T A G T A G C T A A T T T T G C
A A G C T A G T A G C T A A T T T T G C
A A G C T N G T A G C T A A T T T T G C
A T G T T G G T A G C T A G T T T A G C
```



- **Reconstructs haplotypes:** Locally reassembles reads into candidate sequences rather than examining sites independently.
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```
ATGTTGGTAGCTAGTTTAGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
ATGTTGGTAGCTAGTTTAGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
ATGTTGGTAGCTAGTTTAGC
ATGTTGGTAGCTAGTTTAGC
ATGTTGGTAGCTAGTTTAGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
ATGTTGGTAGCTAGTTTAGC
```

OPEN

## The evaluation of Bcftools mpileup and GATK HaplotypeCaller for variant calling in non-human species

Messaoud Lefouili & Kiwoong Nam 

Identification of genetic variations is a central part of population and quantitative genomics studies based on high-throughput sequencing data. Even though popular variant callers such as Bcftools mpileup and GATK HaplotypeCaller were developed nearly 10 years ago, their performance is still largely unknown for non-human species. Here, we showed by benchmark analyses with a simulated insect population that Bcftools mpileup performs better than GATK HaplotypeCaller in terms of recovery rate and accuracy regardless of mapping software. The vast majority of false positives were observed from repeats, especially for GATK HaplotypeCaller. Variant scores calculated by GATK did not clearly distinguish true positives from false positives in the vast majority of cases, implying that hard-filtering with GATK could be challenging. These results suggest that Bcftools mpileup may be the first choice for non-human studies and that variants within repeats might have to be excluded for downstream analyses.

## **1.4 Filtering variants for quality**

# You now have genotypes! Job done?

Not quite! many of these may be unreliable due to poor sequencing coverage and others will be artifacts of thing like poor alignment

| #CHROM  | POS  | Ind1 | Ind2 | Ind3 | Ind4 | Ind5 | Ind6 |
|---------|------|------|------|------|------|------|------|
| Contig1 | 100  | 0/0  | 0/0  | 0/1  | 0/1  | 1/1  | 1/1  |
| Contig1 | 200  | 0/0  | 0/1  | 0/1  | 0/0  | 0/1  | 1/1  |
| Contig1 | 300  | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  | 1/1  |
| Contig1 | 400  | 0/0  | 0/0  | 0/0  | 0/1  | 0/1  | 0/1  |
| Contig1 | 500  | 0/1  | 0/1  | 1/1  | 0/0  | 0/1  | 1/1  |
| Contig1 | 600  | 0/0  | 0/1  | 0/0  | 1/1  | 0/1  | 1/1  |
| Contig1 | 700  | 1/1  | 1/1  | 0/1  | 0/1  | 0/0  | 0/0  |
| Contig1 | 800  | 0/0  | 0/0  | 0/1  | 0/1  | 0/1  | 1/1  |
| Contig1 | 900  | 0/1  | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  |
| Contig1 | 1000 | 0/0  | 0/1  | 1/1  | 0/1  | 0/0  | 1/1  |

# Two ways we can apply filters

**Hard Filtering:** The removal of entire *sites* from a vcf file

| #CHROM  | POS  | Ind1 | Ind2 | Ind3 | Ind4 | Ind5 | Ind6 |
|---------|------|------|------|------|------|------|------|
| Contig1 | 100  | 0/0  | 0/0  | 0/1  | 0/1  | 1/1  | 1/1  |
| Contig1 | 200  | 0/0  | 0/1  | 0/1  | 0/0  | 0/1  | 1/1  |
| Contig1 | 300  | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  | 1/1  |
| Contig1 | 400  | 0/0  | 0/0  | 0/0  | 0/1  | 0/1  | 0/1  |
| Contig1 | 500  | 0/1  | 0/1  | 1/1  | 0/0  | 0/1  | 1/1  |
| Contig1 | 600  | 0/0  | 0/1  | 0/0  | 1/1  | 0/1  | 1/1  |
| Contig1 | 700  | 1/1  | 1/1  | 0/1  | 0/1  | 0/0  | 0/0  |
| Contig1 | 800  | 0/0  | 0/0  | 0/1  | 0/1  | 0/1  | 1/1  |
| Contig1 | 900  | 0/1  | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  |
| Contig1 | 1000 | 0/0  | 0/1  | 1/1  | 0/1  | 0/0  | 1/1  |

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| Contig1 | 100  | 0/0  | 0/0  | 0/1  | 0/1  | 1/1  | 1/1  |
| Contig1 | 300  | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  | 1/1  |
| Contig1 | 400  | 0/0  | 0/0  | 0/0  | 0/1  | 0/1  | 0/1  |
| Contig1 | 500  | 0/1  | 0/1  | 1/1  | 0/0  | 0/1  | 1/1  |
| Contig1 | 600  | 0/0  | 0/1  | 0/0  | 1/1  | 0/1  | 1/1  |
| Contig1 | 800  | 0/0  | 0/0  | 0/1  | 0/1  | 0/1  | 1/1  |
| Contig1 | 1000 | 0/0  | 0/1  | 1/1  | 0/1  | 0/0  | 1/1  |

# Two ways we can apply filters

**Soft filtering of genotypes:** The masking of *genotypes* from a VCF file

| #CHROM  | POS  | Ind1 | Ind2 | Ind3 | Ind4 | Ind5 | Ind6 |
|---------|------|------|------|------|------|------|------|
| Contig1 | 100  | 0/0  | 0/0  | 0/1  | 0/1  | 1/1  | 1/1  |
| Contig1 | 200  | 0/0  | 0/1  | 0/1  | 0/0  | 0/1  | 1/1  |
| Contig1 | 300  | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  | 1/1  |
| Contig1 | 400  | 0/0  | 0/0  | 0/0  | 0/1  | 0/1  | 0/1  |
| Contig1 | 500  | 0/1  | 0/1  | 1/1  | 0/0  | 0/1  | 1/1  |
| Contig1 | 600  | 0/0  | 0/1  | 0/0  | 1/1  | 0/1  | 1/1  |
| Contig1 | 700  | 1/1  | 1/1  | 0/1  | 0/1  | 0/0  | 0/0  |
| Contig1 | 800  | 0/0  | 0/0  | 0/1  | 0/1  | 0/1  | 1/1  |
| Contig1 | 900  | 0/1  | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  |
| Contig1 | 1000 | 0/0  | 0/1  | 1/1  | 0/1  | 0/0  | 1/1  |

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|---------|------|------|------|------|------|------|------|
| Contig1 | 100  | 0/0  | 0/0  | 0/1  | 0/1  | 1/1  | 1/1  |
| Contig1 | 200  | 0/0  | ./.  | 0/1  | 0/0  | 0/1  | ./.  |
| Contig1 | 300  | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  | 1/1  |
| Contig1 | 400  | 0/0  | 0/0  | 0/0  | 0/1  | 0/1  | 0/1  |
| Contig1 | 500  | 0/1  | 0/1  | ./.  | 0/0  | 0/1  | 1/1  |
| Contig1 | 600  | 0/0  | 0/1  | 0/0  | 1/1  | 0/1  | 1/1  |
| Contig1 | 700  | 1/1  | 1/1  | 0/1  | 0/1  | 0/0  | 0/0  |
| Contig1 | 800  | 0/0  | 0/0  | 0/1  | 0/1  | ./.  | 1/1  |
| Contig1 | 900  | 0/1  | 1/1  | 0/1  | 0/0  | 0/0  | 0/1  |
| Contig1 | 1000 | 0/0  | 0/1  | 1/1  | 0/1  | 0/0  | 1/1  |

## Common hard filters: **Max depth** (INFO/DP)

If we sequence to an expected depth of  $x$ , we expect some variation across sites around  $x$  due simply to chance.

Unusually high depth can indicate a problem!

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If we sequence to an expected depth of  $x$ , we expect some variation across sites around  $x$  due simply to chance.

Unusually high depth can indicate a problem!

- **Collapsed repetitive sequences:** several similar genomic regions are represented by one location in the reference, so reads from all copies map there.
- **Duplicated genes or paralogues:** reads from related gene copies incorrectly map to the same reference position.
- **Copy-number variation:** some individuals genuinely carry additional copies of the region.

## Common hard filters: **Variant confidence** (QUAL)

A Phred-scaled confidence that the site has been correctly called

- Score of 30 indicates a 1 in 1,000 change of an error

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A Phred-scaled confidence that the site has been correctly called

- Score of 30 indicates a 1 in 1,000 change of an error

Integrates information about

- **Bases observed:** how many reads support the reference and alternative alleles.
- **Base quality:** whether individual bases are likely to have been sequenced correctly.
- **Mapping quality:** whether the reads are likely to have been aligned correctly
- **Read depth:** more reads supporting an allele generally provide stronger evidence.
- **Alignment uncertainty:** BCFtools mpileup uses alignment info to reduce the influence of bases that may be misaligned, particularly around indels.
- **Genotype likelihoods:** for each individual, it calculates the likelihood of genotypes such as 0/0, 0/1 and 1/1.
- **Evidence across individuals:** BCFtools call combines the genotype likelihoods across samples and applies its population-genetic model to assess whether an alternative allele is present.

## Common hard filters: **Strand Bias** (SP)

A Phred-scaled p-value indicated whether the reference and alternative alleles are supported differently by forward- and reverse-strand reads.

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# Common hard filters: **Strand Bias** (SP)

A Phred-scaled p-value indicated whether the reference and alternative alleles are supported differently by forward- and reverse-strand reads.

- Score of 30 indicates a 1 in 1,000 change of an error

Naively, we expect the forward and reverse strands to be sequenced the same number of times, but bias can be produced:

- **PCR bias:** some fragments may amplify more successfully in one orientation.
- **Alignment artefacts:** reads from one strand may map differently
- **Read-position effects:** the alternative allele may occur near the ends of reads from one orientation, where error rates are higher.
- **Adapter contamination or incomplete trimming:** residual adapter sequence can affect one read orientation more strongly.
- **Library-preparation biases:** fragmentation, amplification or target-capture steps can produce unequal strand representation.

## Soft filters: **Genotype quality** (QG) & **depth** (FORMAT/DP)

Here we are evaluating each genotype based on its individual quality score. For a large VCF this can mean billions of evaluations.

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Here we are evaluating each genotype based on its individual quality score. For a large VCF this can mean billions of evaluations.

**GQ < 30 | DP < 5**

| Sample | Original genotype |
|--------|-------------------|
| A      | 0/1               |
| B      | 0/1               |
| C      | 1/1               |
| D      | 0/0               |

## Soft filters: Genotype quality (GQ) & depth (FORMAT/DP)

Here we are evaluating each genotype based on its individual quality score. For a large VCF this can mean billions of evaluations.

$GQ < 30 \mid DP < 5$

| Sample | Original genotype | GQ | DP | Result |
|--------|-------------------|----|----|--------|
| A      | 0/1               | 35 | 12 | 0/1    |
| B      | 0/1               | 12 | 10 | ./.    |
| C      | 1/1               | 40 | 3  | ./.    |
| D      | 0/0               | 30 | 5  | 0/0    |

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Here we are evaluating each genotype based on its individual quality score. For a large VCF this can mean billions of evaluations.

$GQ < 30 \mid DP < 5$

| Sample | Original genotype | GQ | DP | Result |
|--------|-------------------|----|----|--------|
| A      | 0/1               | 35 | 12 | 0/1    |
| B      | 0/1               | 12 | 10 | ./.    |
| C      | 1/1               | 40 | 3  | ./.    |
| D      | 0/0               | 30 | 5  | 0/0    |

After this step it is important to remove sites with > some % missing data

# How much do we lose during filtering?

| Dataset                 | Total sites   | Total invariant | Total SNPs | Multallele SNPs |
|-------------------------|---------------|-----------------|------------|-----------------|
| Full unfiltered dataset | 1,658,076,957 | 1,521,844,025   | 99,448,272 | 17,240,263      |

# How much do we lose during filtering?

| Dataset                                                                                                         | Total sites   | Total invariant | Total SNPs | Multallele SNPs |
|-----------------------------------------------------------------------------------------------------------------|---------------|-----------------|------------|-----------------|
| <b>Full unfiltered dataset</b>                                                                                  | 1,658,076,957 | 1,521,844,025   | 99,448,272 | 17,240,263      |
| <b>After Hard filters</b><br>-e 'FS>60.0    SOR>3    MQ<40    MQRankSum<-12.5    QD<2.0    ReadPosRankSum<-8.0' | 1,527,806,671 | 1,429,859,187   | 74,224,476 | 8,485,685       |

# How much do we lose during filtering?

| Dataset                                                                                                         | Total sites   | Total invariant | Total SNPs | Multallele SNPs |
|-----------------------------------------------------------------------------------------------------------------|---------------|-----------------|------------|-----------------|
| <b>Full unfiltered dataset</b>                                                                                  | 1,658,076,957 | 1,521,844,025   | 99,448,272 | 17,240,263      |
| <b>After Hard filters</b><br>-e 'FS>60.0    SOR>3    MQ<40    MQRankSum<-12.5    QD<2.0    ReadPosRankSum<-8.0' | 1,527,806,671 | 1,429,859,187   | 74,224,476 | 8,485,685       |
| <b>After Drop SNPs near indels &amp; complex vars</b><br>-g 5:indel,other                                       | 1,499,745,457 | 1,429,859,176   | 46,163,273 | 2,620,532       |

# How much do we lose during filtering?

| Dataset                                                                                                         | Total sites   | Total invariant | Total SNPs | Multallele SNPs |
|-----------------------------------------------------------------------------------------------------------------|---------------|-----------------|------------|-----------------|
| <b>Full unfiltered dataset</b>                                                                                  | 1,658,076,957 | 1,521,844,025   | 99,448,272 | 17,240,263      |
| <b>After Hard filters</b><br>-e 'FS>60.0    SOR>3    MQ<40    MQRankSum<-12.5    QD<2.0    ReadPosRankSum<-8.0' | 1,527,806,671 | 1,429,859,187   | 74,224,476 | 8,485,685       |
| <b>After Drop SNPs near indels &amp; complex vars</b><br>-g 5:indel,other                                       | 1,499,745,457 | 1,429,859,176   | 46,163,273 | 2,620,532       |
| <b>After retaining only bialleleic SNPs</b><br>--exclude-types indels,other --max-alleles 2                     | 1,473,401,917 | 1,429,859,176   | 43,542,741 | -               |

# How much do we lose during filtering?

| Dataset                                                                                                         | Total sites   | Total invariant | Total SNPs | Multallele SNPs |
|-----------------------------------------------------------------------------------------------------------------|---------------|-----------------|------------|-----------------|
| <b>Full unfiltered dataset</b>                                                                                  | 1,658,076,957 | 1,521,844,025   | 99,448,272 | 17,240,263      |
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| <b>After Drop SNPs near indels &amp; complex vars</b><br>-g 5:indel,other                                       | 1,499,745,457 | 1,429,859,176   | 46,163,273 | 2,620,532       |
| <b>After retaining only bialleleic SNPs</b><br>--exclude-types indels,other --max-alleles 2                     | 1,473,401,917 | 1,429,859,176   | 43,542,741 | -               |
| <b>After dropping sites above threshold DP</b><br>-e 'INFO/DP>3750'                                             | 1,450,599,180 | 1,408,460,066   | 42,139,114 | -               |

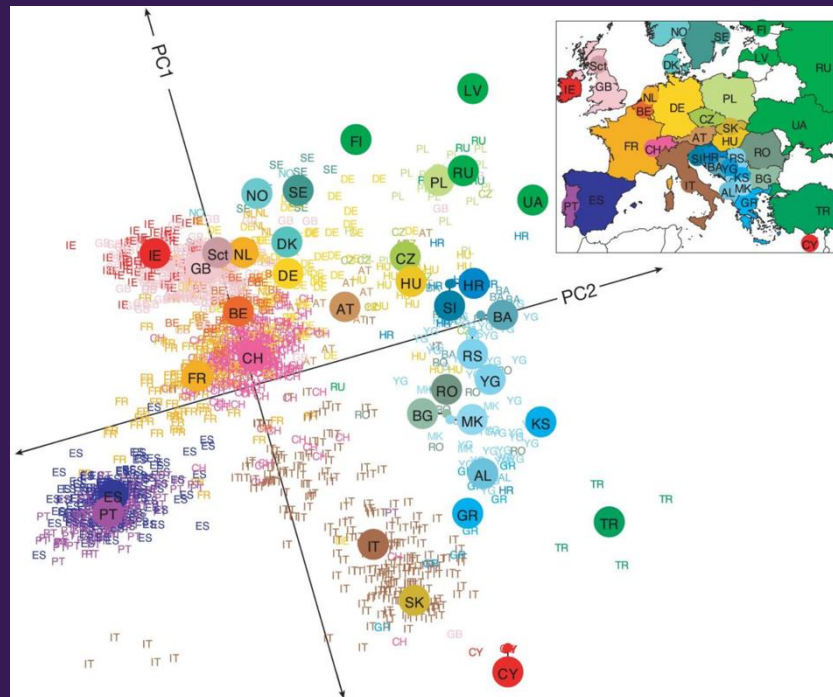
# How much do we lose during filtering?

| Dataset                                                                                                                                                  | Total sites   | Total invariant | Total SNPs | Multallele SNPs |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------|------------|-----------------|
| <b>Full unfiltered dataset</b>                                                                                                                           | 1,658,076,957 | 1,521,844,025   | 99,448,272 | 17,240,263      |
| <b>After Hard filters</b><br>-e 'FS>60.0    SOR>3    MQ<40    MQRankSum<-12.5    QD<2.0    ReadPosRankSum<-8.0'                                          | 1,527,806,671 | 1,429,859,187   | 74,224,476 | 8,485,685       |
| <b>After Drop SNPs near indels &amp; complex vars</b><br>-g 5:indel,other                                                                                | 1,499,745,457 | 1,429,859,176   | 46,163,273 | 2,620,532       |
| <b>After retaining only bialleleic SNPs</b><br>--exclude-types indels,other --max-alleles 2                                                              | 1,473,401,917 | 1,429,859,176   | 43,542,741 | -               |
| <b>After dropping sites above threshold DP</b><br>-e 'INFO/DP>3750'                                                                                      | 1,450,599,180 | 1,408,460,066   | 42,139,114 | -               |
| <b>After soft filtering:</b><br><b>Set low DP genotypes to missing</b><br>-S . -e 'FMT/DP<5   FMT/GQ<20'<br><b>drop sites with &gt; 0.2 missing data</b> | 454,361,480   | 429,399,898     | 24,961,582 | -               |

# How much do we lose during filtering?

| Dataset                                                                                                                                                                                                                                             | Total sites   | Total invariant | Total SNPs | Multallele SNPs |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------|------------|-----------------|
| <b>Full unfiltered dataset</b>                                                                                                                                                                                                                      | 1,658,076,957 | 1,521,844,025   | 99,448,272 | 17,240,263      |
| <b>After Hard filters</b><br>-e 'FS>60.0    SOR>3    MQ<40    MQRankSum<-12.5    QD<2.0    ReadPosRankSum<-8.0'                                                                                                                                     | 1,527,806,671 | 1,429,859,187   | 74,224,476 | 8,485,685       |
| <b>After Drop SNPs near indels &amp; complex vars</b><br>-g 5:indel,other                                                                                                                                                                           | 1,499,745,457 | 1,429,859,176   | 46,163,273 | 2,620,532       |
| <b>After retaining only bialleleic SNPs</b><br>--exclude-types indels,other --max-alleles 2                                                                                                                                                         | 1,473,401,917 | 1,429,859,176   | 43,542,741 | -               |
| <b>After dropping sites above threshold DP</b><br>-e 'INFO/DP>3750'                                                                                                                                                                                 | 1,450,599,180 | 1,408,460,066   | 42,139,114 | -               |
| <b>After soft filtering:</b><br><b>Set low DP genotypes to missing</b><br>-S . -e 'FMT/DP<5   FMT/GQ<20'<br><b>drop sites with &gt; 0.2 missing data</b><br>-e 'F_MISSING > 0.2'                                                                    | 454,361,480   | 429,399,898     | 24,961,582 | -               |
| <b>After additional hard and soft filtering</b><br><b>Drop sites below depth threshold</b><br>-e 'INFO/DP<1500'<br><b>Drop genotypes with &lt; 10 reads</b><br>-e 'FMT/DP<10',<br><b>Drop sites with &gt; 10% missing data</b><br>'F_MISSING > 0.1' | 328,831,939   | 310,294,741     | 18,537,198 | -               |

## 2.1 Principal components analysis



# PCA can be used to summarise genomic datasets

**The first analysis that most people do!**

- Each SNP represents a dimension of genetic variation.
- With millions of SNPs, individuals occupy a high-dimensional “genetic space”.
- PCA reduces this variation to a few axes called principal components (PCs).

# How does it do?

- PCA identifies combinations of SNPs whose allele counts covary across individuals.
- For example, if individuals carrying allele A at one SNP also tend to carry particular alleles at many other SNPs, those SNPs may contribute in the same direction to a principal component.
- PCA combines these correlated allele-frequency patterns into axes.

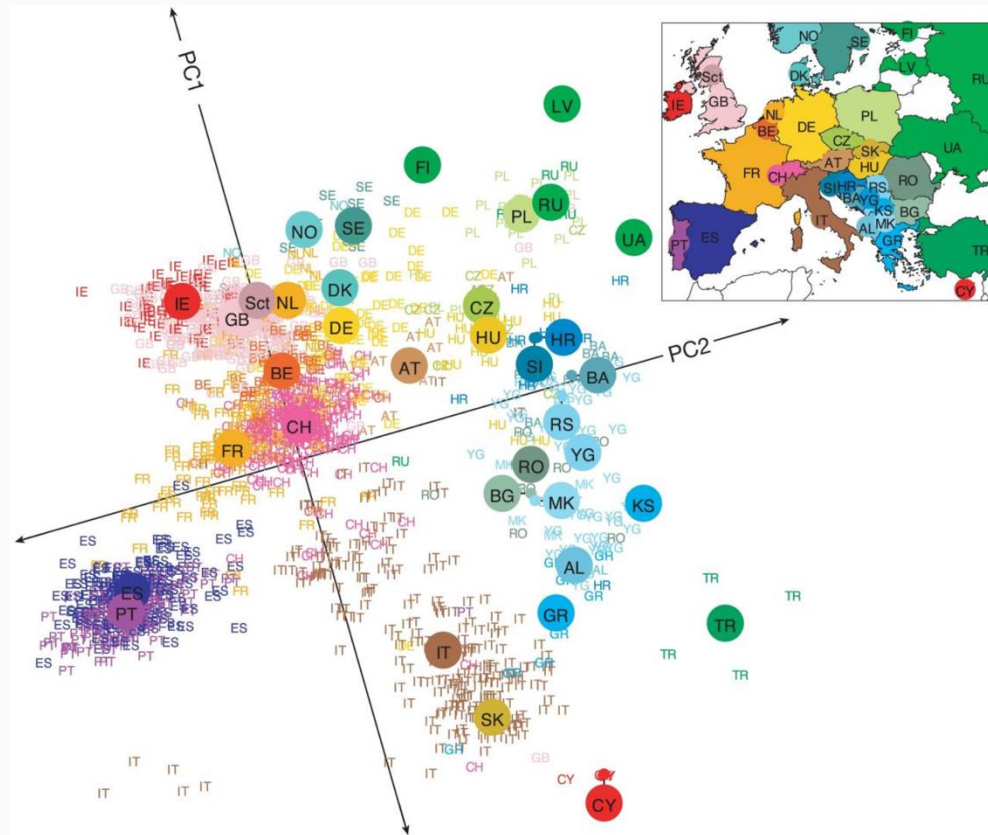
# PCA can be used to summarise genomic datasets

## PCA can reveal

- Major patterns of population structure
- Genetic clusters and geographic gradients
- Admixed or intermediate individuals
- Closely related, duplicated or unusual samples
- Sample mislabelling and technical batch effects

# Some examples

## Genes mirror geography within Europe



3,000 European individuals genotyped at over half a million variable DNA sites in the human genome.

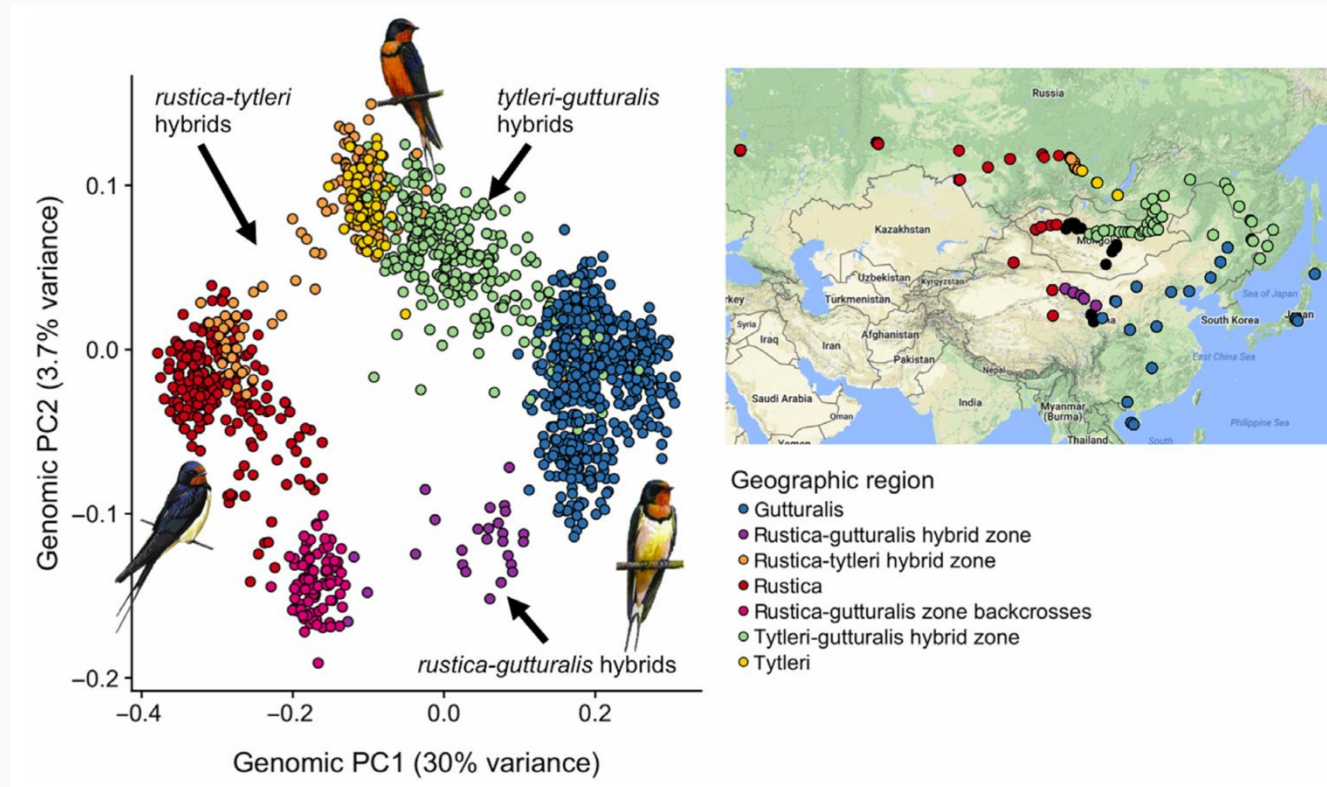
Despite low average levels of genetic differentiation among Europeans, we find a close correspondence between genetic and geographic distances

indeed, a geographical map of Europe arises naturally as an efficient two-dimensional summary of genetic variation in Europeans.

<https://www.nature.com/articles/nature07331>

# Some examples

## A ring around the Tibetan Plateau

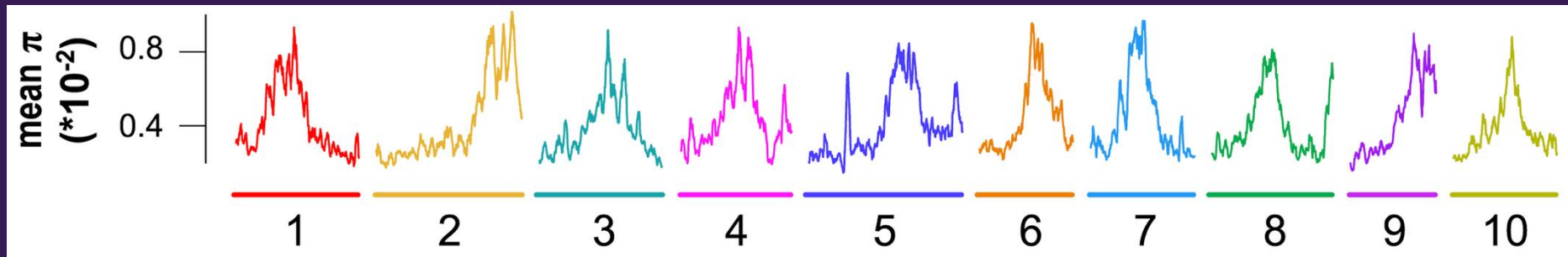


We identified three genetic clusters that corresponded to the three subspecies and varied dramatically in extent of admixture.

We found limited hybridisation between *rustica tytleri* and *rustic gutturalis*, but extensive admixture between *tytleri* and *gutturalis*

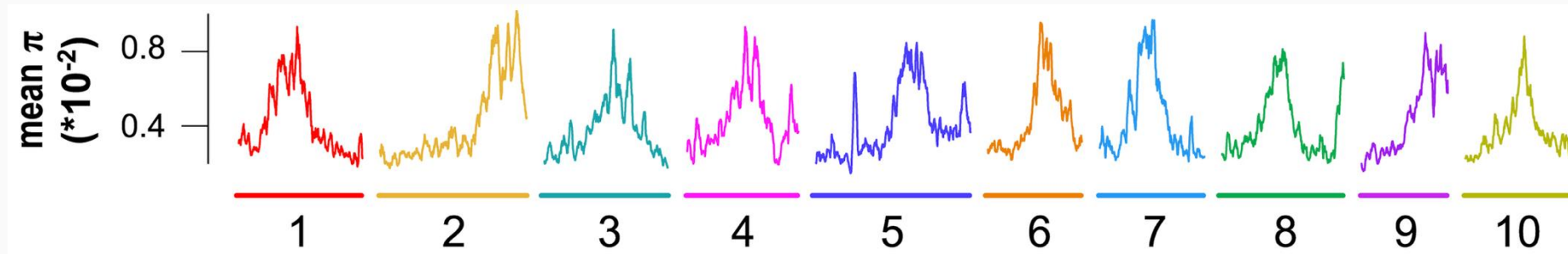
<https://onlinelibrary.wiley.com/doi/full/10.1111/ele.13420>

## 2.2 Genome scans



# Genome scans

- A genome scan examines how a genetic statistic varies along the genome.
- Statistics are calculated repeatedly across chromosomes or contigs.
- Peaks and troughs identify regions that differ from the genomic background.
- Genome scans can reveal candidate regions associated with:
  - Selection
  - Reduced gene flow
  - Recombination-rate variation
  - Structural variants
  - Unusually high or low genetic diversity

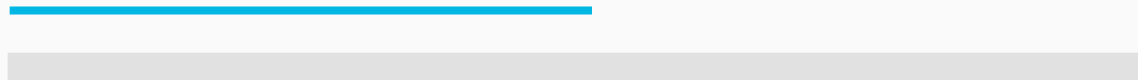


# Why calculate statistics in windows?

- Individual SNPs contain limited and noisy information.
- Genome scans group neighbouring sites into **windows**.
- Larger windows contain more information but give lower spatial resolution.
- Smaller windows give higher resolution but produce noisier estimates.
- Windows can contain a fixed number of base pairs or a fixed number of SNPs.

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**Window size:** the amount of sequence included in each estimate

**Step size:** the distance the window moves before the next estimate

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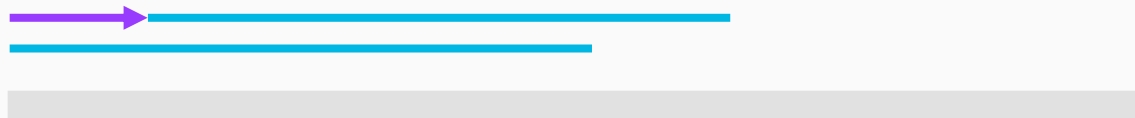


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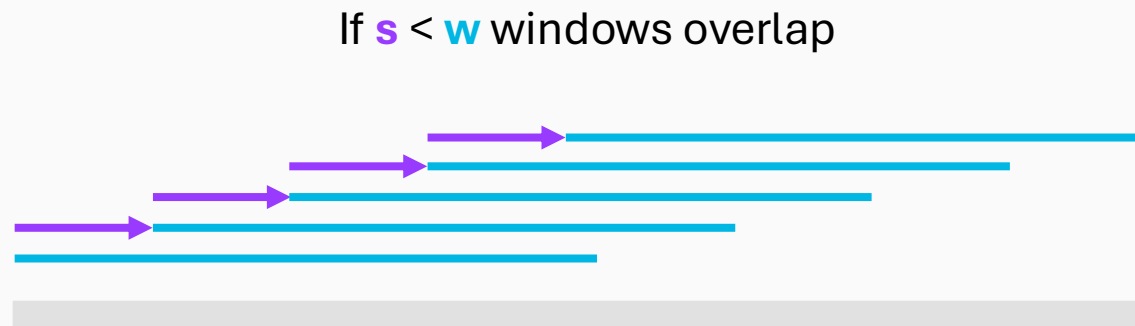


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If  $s = w$  windows are non-overlapping



**Window size:** the amount of sequence included in each estimate

**Step size:** the distance the window moves before the next estimate

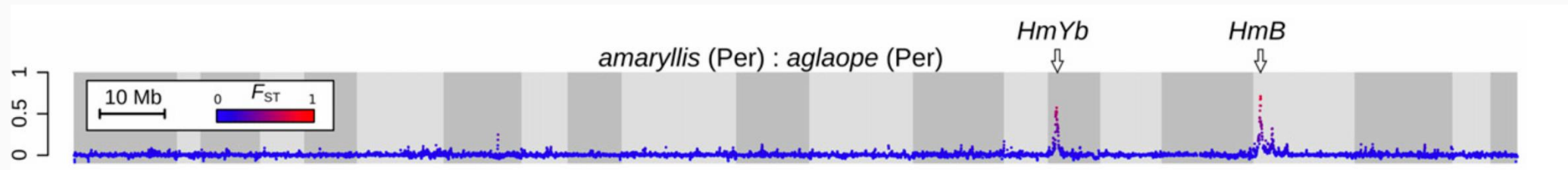
# What statistics can be calculated in windows?

- Anything you want!
- We will calculate classic measures of within- and between-population variation

| Statistic        | What does it measure?                                               | Interpretation              |
|------------------|---------------------------------------------------------------------|-----------------------------|
| $\pi$            | Mean pairwise difference between sequences within a population      | Genetic diversity           |
| $d_{XY} (\pi_b)$ | Mean pairwise difference between sequences from populations X and Y | Absolute genetic divergence |
| $F_{ST}$         | Differentiation relative to the diversity within populations        | Relative genetic divergence |

# An example

## Genome-wide evidence for speciation with gene flow in *Heliconius* butterflies

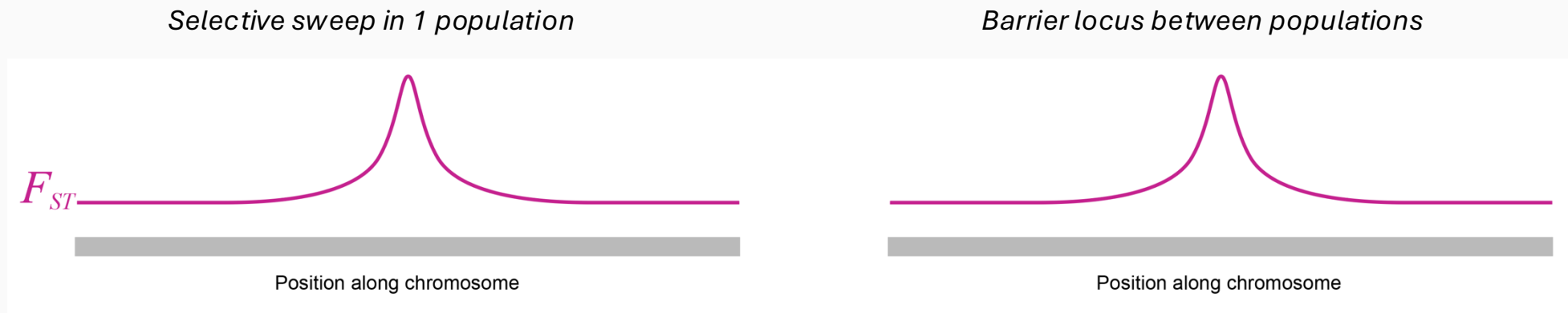


We characterized patterns of divergence across the genome between populations at various levels of divergence and geographic separation using the fixation index,  $F_{ST}$ . At the earliest stage of divergence, between parapatric races,  $F_{ST}$  was low throughout the genome with just a few narrow peaks, which are partly explained by known wing pattern divergence. Between *H. m. aglaope* and *H. m. amaryllis* from Peru

<https://genome.cshlp.org/content/23/11/1817>

# Interpret genome scans with caution!

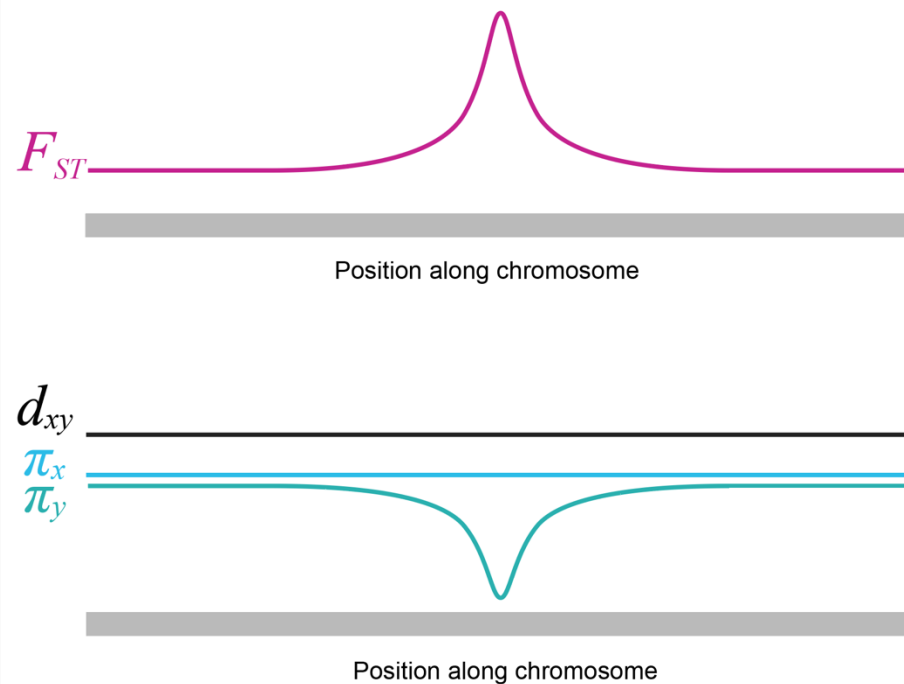
- $F_{ST}$  is a ratio of within to between population diversity i.e.,  $F_{ST} \approx 1 - (\pi/d_{xy})$
- This means that  $F_{ST}$  can go up and down for different reasons



# Interpret genome scans with caution!

- $F_{ST}$  is a ratio of within to between population diversity i.e.,  $F_{ST} \approx 1 - (\pi/d_{xy})$
- This means that  $F_{ST}$  can go up and down for different reasons

*Selective sweep in 1 population*



*Barrier locus between populations*

